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Preserving Scientists During Wars and Emergencies

White Paper

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Project: Science at Risk

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Executive summary

This publication was prepared for the Science at Risk project, which aims to disseminate knowledge about science at risk, including the functioning of Ukrainian science and scientists during Russia's full-scale war against Ukraine. Ukrainian science has been losing human potential for a long period due to the unattractiveness of scientific careers for the youth, leading to an increased median age among scientists. Russia's full-scale invasion has dramatically aggravated this issue.

The principal threats to preserving human resources in science during armed conflicts include emigration, direct casualties and war-related traumas, and professional mobility outside the scientific sector. This White Paper aims to investigate the principal risks and threats of losing scientists due to the war between Russia and Ukraine, examine their experience conducting research during the war, and propose mechanisms for preserving scientists and scientific potential during martial law.

People are the most vulnerable in crises and upheavals, focusing on satisfying basic needs and survival. In every armed conflict, people (scientists) suffer, facing a direct threat to their lives and well-being, restrictions on conducting research, and forced relocation. It leads to migrational flows of refugees, disrupting professional networks and connections and excluding young, promising researchers from science. Scientists face violations of fundamental human rights (right to life, inviolability of the home, etc.) and restrictions on academic freedom, making research impossible.

Ukraine doesn't have a unified system for tracking losses in scientific human resources. The available data on the detrimental impact of war on Ukrainian scientists is fragmented. As of 07.20/2023, 2,202 scientific officers from National Academy of Sciences of Ukraine (NASU) institutions (16% of the total) were unable to work at their facilities. Of these, 1,497 (11%) emigrated abroad, while 705 (5%) were internally displaced persons within the country. The total number of scientists who left the country exceeds 6,000.

The National Research Foundation reports that under martial law, only 57 of 169 teams are ready to continue their research and developments that were put on hold, 62 teams may continue work under certain

circumstances, and 50 collectives won't be able to continue with their research at all. One of the reasons for putting research projects on hold is funding interruptions.

Among the top 10 countries Ukrainian scientists traditionally collaborated with are Poland, Germany, the USA, Great Britain, China, France, Italy, the Czech Republic, Greece, Slovakia, and Russia. In 2020-2021, Russia held second place on this list. The full-scale invasion rapidly limited the collaboration between scientists from Ukraine and Russia: by 2022, Russia descended to the fourth position in scientific collaboration, and by 2023, it dropped to the eleventh place.

Analysis of publication activity in scientific journals of the MDPI publisher reveals a decrease in international collaborations involving authors with Ukrainian affiliation during 2019-2022 with all countries except Poland.

International partners have taken various measures to support Ukrainian scientists, including expanded support for Ukrainian projects in Erasmus+ competitions and programs. Invaluable aid was granted to Ukrainians by national research agencies of many countries and international research networks. The powerful international support has allowed the scientists (particularly women) to relocate abroad and continue their research. The scientists who found their temporary refuge abroad and are continuing their research are creating a new generation of Ukrainian diaspora that can conduct efficient scientific diplomacy and contribute to developing international scientific partnerships.

As a result of this research, we propose the following recommendations:

- Establish a centralized system for tracking data on the losses of scientists during the war, using the Ministry of Education and Science (MES) as the foundation, with regular and systematic publication of real-time information. It will give a clear understanding of the overall risks, threats, and destructive consequences of Russia's full-scale invasion for science in general and scientific human potential in particular.
- Restructure and develop existing programs to support scientists who remain in Ukraine or have returned from abroad (remote vacancies in foreign institutions, virtual mobility and remote grants, enlisting researchers and research groups through outsourcing, creating new collaborative networks, etc.).
- Regulate the issues of remote work for scientists during the war. Despite there being no legal prohibitions for remote work, there is a gap in regulations that would protect scientists who found a temporary refuge abroad and can potentially lose connections with their institutions or cease conducting research on their topics due to the pressure from the management or forced quitting.
- Create and develop support programs for scientists actively serving on the frontlines and those returning to work after serving in the Armed Forces of Ukraine (AFU): adaptation within scientific collectives, integration support for returning to research topics, development of policies for barrier-free access to the creation of scientific outputs, psychological support and mental well-being programs, inclusivity.

Introduction

Throughout history, wars and social upheavals burst out unexpectedly, posing the most complicated challenges. Research of informational sources on armed conflicts allows us to distinguish three aspects of their impact on science: 1) loss of human capital; 2) destruction of infrastructure; and 3) decline in international collaboration. Research shows that restoring infrastructure or physical capital is much easier and faster than human capital. The loss of human capital in science leads to long-term catastrophic losses. Despite this, the ruining of infrastructure is easier to estimate, and therefore, the focus is shifted to it. Amid the chaos and destruction of armed conflicts, protection and development of human capital in science are often overlooked, while the necessity to protect civilians, solve humanitarian issues, provide law enforcement, and give medical aid comes to the fore. History shows that war can serve both as a catalyst and an obstacle to scientific progress. Wars and upheavals can cause quality changes in society. Consequently, preserving scientists and research groups is crucial as they are highly vulnerable in this period. Our work focuses on the research on Ukrainian scientists and their studies during Russia's full-scale invasion of Ukraine and encompasses the period from 2022 to 2023.

The war in Ukraine has reshaped global scientific partnerships[1] — in particular, giving a start to severing ties with Russian scientists. The uniqueness of the Ukrainian situation is that, for the first time since World War II, (1) the frontline spans 2,450 km, with active combat occurring along 1,105 km;[2] (2) technologies in warfare have changed with a focus on informational technologies, robotics, and drones; (3) there is unprecedented terror and genocide of the civilians, evidenced by total deurbanization and destruction of cities and villages, hospitals and educational institutions, and objects of communal and civilian infrastructure; (4) there is a global threat to the existence of humanity in the form of

[1] Gaind, N., Abbott, A., Witze, A., Gibney, E., Tollefson, J., Irwin, A., & Van Noorden, R. (2022, July 20). Seven ways the war in Ukraine is changing global science. Nature News. *https://www.nature.com/articles/d41586-022-01960-0#*.

[2] As of 06/22/2023, the announcement by the Commander-in-Chief of UAF V. Zaluzhny. URL: *https://www.facebook.com/100068564836091/posts/pfbid02UpKf6VPKcsFy8ZDGGp2e6FCdJxeNLABLy1HncMUrBPkrKeYJhY38Mh2sTCrw4W5Wl/.*

nuclear blackmail, including the seizure of Europe's largest nuclear power plant.

Ukraine has a powerful scientific potential, evident in its stand in the Global Innovation Index in 2023, a year of countering the Russian full-scale invasion[3]: Ukraine ranks 55th out of 139 world economies overall, with the third position among 37 economies with low and middle-income and holds the 34th position among 39 European economies. In such circumstances, the Ukrainian scientific community remains actively engaged in scientific work, organizing scientific events, and participating in the educational process. This is recognized by awards from the world scientific community, including the Emma Goldman Award (K. Zaremba, T. Martseniuk), the Fields Medal (Maryna Viazovska), etc. The achievements and functioning of Ukrainian science during the war can serve as a vivid example of the resilience and endurance of scientists in crises; the experiences of Ukrainian scientists in preserving scientific potential, understanding the primary risks, and looking for ways to overcome them can offer valuable insights for the international community and the future development of policy in the safety of human capital.

This research aims to investigate the primary risks and threats of losing scientists due to Russia's war in Ukraine, as well as mechanisms of documenting losses in scientific potential resulting from casualties, emigration of scientists seeking refuge abroad, and research termination due to service in the Armed Forces of Ukraine (AFU). According to the goal of the research, the following tasks were defined:

- study the international experience of armed conflicts impact on scientists and their ability to conduct research;
- understand the mechanism for documenting losses in science and among scientists in wartime in Ukraine;
- assess the war's impact on producing scientific results, including scientific partnerships;
- study the experience of the Ukrainian scientists conducting research during wartime;
- offer mechanisms to safeguard scientists and scientific potential under martial law.

This paper draws on the research on preserving scientists and scientific potential during other armed conflicts, interviews, and the results of the survey conducted among Ukrainian scientists during 2022-2023, detailed further in the Methodology and Structure of the Research section.

The principal threats to preserving human resources in science during armed conflicts include emigration, direct casualties and war-related traumas, and professional mobility outside the scientific sector. Studying the Ukrainian experience is important for the global community to understand not only how to preserve human capital during armed military conflicts but also how to maintain scientific research and create new knowledge for progress and development.

[3] Global Innovation Index 2023. URL: *https://www.wipo.int/global_innovation_index/en/2023/?utm_source=google&utm_medium=cpc&utm_campaign=Search%3A+GIobal+Innovation+Index+2023+%28EN%29&gclid=Cj0KCQjwm66pBhDQARIsALIR2zCLtaYUbK61k_2f9Dx4P3mj_vacfWwhfOOiv3asLAN-YA2xf41IBPcaAm05EALw_wcB.*

Methodology and Structure of the Research

Our work on this topic started with desk research, involving an analysis of scientific publications, contemporary statistical information, and data from open sources pertaining to the preservation of scientists during wars and emergencies. Encyclopedia Britannica defines a scientist as "a person who is trained in a science and whose job involves doing scientific research or solving scientific problems.[4]" Ukrainian law defines a scientist as an individual engaged in fundamental and/or applied scientific research who obtains scientific and/or scientific and technological (applied) results[5]. The Collins dictionary defines "scientists" and "researcher" as synonyms; therefore, this text uses these terms interchangeably[6]. In a broad international context[7] **a scientist (researcher, scholar) at risk** is defined as a person facing threats, persecution, or danger due to their work or activities related to research, innovation, or other academic functions. These threats may arise for different reasons, including armed conflicts, climate cataclysms, or restrictions on academic freedom imposed by authoritarian political regimes. In this research, the category **"scientist at risk"** is defined as a person engaged in fundamental and/or applied scientific research and obtaining scientific results while facing threats to their life, well-being, and ability to conduct scientific research because of Russia's full-scale invasion of Ukraine.

In times of armed conflicts, the preservation of science can be ensured through 1) the preservation of scientists and their ability to conduct scientific research (preservation of human capital); 2) the preservation of research groups and transfer of scientific knowledge (preservation of scientific potential and personnel training); 3) the preservation of scientific topics of Ukrainian scientists in the global scientific landscape. This research focuses on identifying ways to preserve and ensure the ability to conduct research for scientists at risk.

[4] Encyclopædia Britannica, inc. (n.d.). scientist. Encyclopædia Britannica. URL: *https://www.britannica.com/dictionary/scientist#*.
[5] Law of Ukraine #848-VIII from 09/03/2023 "About Scientific and Technological and Scientific Work". URL: *https://zakon.rada.gov.ua/laws/show/848-19#*.
[6] *https://www.collinsdictionary.com/dictionary/english-thesaurus/scientist*
[7] Based on materials of the resource *https://www.scholarsatrisk.org/annual-report-2022/*.

In the first stage, we studied the experience and insights gained from previous military conflicts or societal and political upheavals where scientists were at risk and/or faced threats to scientific work (for example, the Taliban's coming to power in Afghanistan).

During the analysis of the impact of Russia's full-scale invasion on Ukrainian science, our primary focus was on analyzing human losses, internal and external mobility, and, in particular, the presence of volunteers or individuals mobilized to the Armed Forces of Ukraine (AFU) or the Territorial Defence Forces, etc. This analysis involved a study of amendments to the laws and legislative initiatives, the data received from the Ministry of Education and Science of Ukraine (MES of Ukraine) and the National Academy of Sciences of Ukraine (NASU), and interviews with representatives of the scientific community.

For this research, we conducted ten in-depth semi-structured interviews and an online survey involving Ukrainian scientists who remain in Ukraine or are abroad (within the UAScience.Reload initiative — the third wave of surveys). The respondents included professionals from MES of Ukraine and NASU, researchers from different scientific fields, managers of international collaboration programs, and rectors of universities of different disciplines and locations. The interviews aimed to discover the experience of preserving scientists during the war and collect expert recommendations. The interviews lasted 1-1,5 hours and were recorded both in audio and video format. As a result, we outlined the main theses and quantitative data (if there were such) on losses of scientific potential during the war in Ukraine, insights on conducting research during crises and possible measures for its preservation, directions for action, and recommendations on every type of loss.

We also used the data from the two waves of UAScience.Reload surveys on the relocation of scientists, issues of financial instability, ability to conduct research, and access to resources.

Considering that it takes it takes 18-26 months to register an invention in Ukraine and 8-12 months to register the utility, we chose the publication activity of Ukrainian scientists reflected in the international scientometric database Scopus, including collaborations with foreign researchers, as the primary indicator for analyzing the impact of the war on the production of scientific results. We based this decision on the fact that despite the extended peer-review procedure for scientific papers, the researcher obtains these public results of their scientific creativity sooner than patents for inventions or utility[8] models. Additionally, patents are more commonly associated with engineering and natural sciences rather than humanities or social sciences.

To summarize the experience of Ukrainian researchers conducting research during the war, we utilized interviews and a focus group. For an in-depth exploration of the motivation/motives behind scientists relocating or returning to Ukraine, we conducted a focus group with nine participants. The questionnaire, organized into four blocks (see Appendix), addressed the following areas:

- motives for relocating abroad;
- barriers to relocating abroad;
- motives for returning to the country to conduct research (What would prompt the scientists to return to their country after a long time abroad?);
- barriers to returning to the country to conduct research (What will be the barrier to returning to Ukraine after researching abroad?).

[8] Law of Ukraine #3687-XII from 04/15/2023 "About Protection of the Invention and Utility Model Rights". URL: *https://zakon.rada.gov.ua/cgi-bin/laws/main.cgi?nreg=3687-12&p=1227171106056773#Text*.

The focus group included Ukrainian scientists who:

- are currently abroad, continuing research with a changed topic;
- are abroad and continue their research and educational work in Ukrainian universities and scientific institutions;
- are abroad but had to cease conducting research;
- were abroad, returned to Ukraine, and continue research in Ukrainian institutions;
- stayed in Ukraine throughout the whole researched period.

During the focus group sessions, special attention was paid to the motivation and reasons to continue conducting research for scientists forced to seek temporary refuge outside of Ukraine. Preserving the scientific potential of researchers currently in AFU and supporting their return to science are also important concerns.

To develop measures for preserving human resources, we used the methods of SWOT analysis and scenario forecasting.

The weaknesses of this research include insufficient attention to vulnerable groups of scientists, such as female scientists with many children, male and female scientists who raise children alone, and scientists who return to scientific work from the Armed Forces of Ukraine (AFU). This is due to the small number of individuals in the last category at the time of the research.

As a result of this research, we draw conclusions and propose recommendations for various groups of stakeholders: 1) heads of scientific institutions and university rectors; 2) representatives of state and local authorities; 3) foreign partners and institutions; 4) active citizens and NGOs.

Chapter I. The Impact of Armed Conflicts of the 20th — 21st Centuries on Scientists and Their Ability to Conduct Research

Attacks on academic freedom are frequent, occur in all parts of the world, and have extensive, sometimes lethal, consequences for scientists, students, and society at large. These attacks occur in closed societies where the right to freedom of thought and expression is continuously suppressed, as well as in cases of political, economic crises and armed conflicts that make researchers especially vulnerable. The World Economic Forum 2023 risk analysis points to the armed conflict in Ukraine and reminds us that 110 similar conflicts are unfolding in the world simultaneously[9]. Throughout the 20th — 21st centuries, numerous conflicts occurred in the world (Table 1), causing the rise of a group of scientists at risk. In most countries where armed conflicts are happening, it's not just the infrastructure that gets ruined; there's a threat to the researchers' lives, and educational institutions are utilized for military purposes.

The research shows that, between January 2005 and March 2015, armed forces or armed groups were utilizing educational facilities in at least 26 countries during conflicts all over Africa, Asia, Europe, the Middle East, and South America[10]. Moreover, educational facilities and research institutions were similarly utilized for military purposes not only by the conflicting parties but also by the peacekeepers. The research[11] particularly emphasizes that groups exposed to the highest risk and suffering in such situations are **women and girls**. In these circumstances, conducting research and training new personnel becomes impossible, leading to losing the country's scientific potential and negatively impacting its progress. At the

[9] Chief risk officers outlook: July 2023. World Economic Forum. URL: *https://www.weforum.org/reports/chief-risk-officers-outlook-july-2023/*.
[10] Lessons in war 2015 – Scholars at risk. URL: *https://www.scholarsatrisk.org/wp-content/uploads/2016/05/Lessons_in_War_2015.pdf*.
[11] Ibid.

same time, history knows examples of when the rapid development of military technology happened thanks to armed conflicts, creating later utilities for civilians.

Armed conflicts mostly burst out unexpectedly and are unpredictable for civilians. For instance, during our work on this research, the war in Israel unfolded abruptly, marked by a significant shelling of Israeli territory by Hamas and the subsequent seizure of districts in the South of Israel. People are most vulnerable in crises and upheavals, focusing on satisfying basic needs and survival. **During every armed conflict, people (scientists) face** a direct threat to their lives and well-being. They encounter prohibitions on conducting research, forced relocations, and disruptions to professional networks, leading to migrational flows of refugees seeking safer places. Emigration waves often have catastrophic and long-term consequences for the home country. One of the primary consequences of researchers' migration is for young scientists — **a disastrous impact on recreating human potential**[12]. For instance, the emigration of Jewish scientists from Nazi Germany resulted in a decline in the number of new Ph.D. students in faculties where high-ranking mathematicians were fired[13]. It shows the relevance of developing methods and algorithms to preserve human resources for post-war scientific recovery. While each conflict's context is unique, there are some fundamental principles and strategies to preserve scientific potential. These include international support, preservation of documentation and archives, support of relocated scientists, and advocacy for academic freedom.

Military conflicts impact the quality of scientific results, influencing both the quantity of publications and shifts in research topics.For example, during World War II there was a notable reduction in fundamental science activities, redirecting efforts toward war-related projects like nuclear physics, electronics, instrumentation, acoustics, and rocket science[14,15]. Wars can lead to increased investments in science and technology, focusing on advancements in military technology or solving war-related problems[16]. Military patronage during the Cold War changed science and technology, making them more project-oriented and subject to national defense restrictions[17].

In 1933, the CARA (the Council for At-Risk Academics)[18] was established to rescue scientists from the Nazi persecution. Ninety years have passed, but researchers worldwide still face discrimination, persecution, suffering, and violence. To safeguard researchers' unique abilities in creating new ideas, innovations, and new knowledge, the international scientific community unites and aids in the name of progress. One such initiative is the global network Scholars at

[12] Ganguli, I., & Waldinger, F. (2023, July 17). War and science in Ukraine. NBER. URL: *https://www.nber.org/papers/w31449*.
[13] Waldinger, F. (2010). Quality Matters: The Expulsion of Professors and the Consequences for PhD Student Outcomes in Nazi Germany. Journal of Political Economy, 118(4), 787–831. URL: *https://doi.org/10.1086/655976*.
[14] Malloryk. (2020, July 31). The scientific and technological advances of World War II: The National WWII Museum: New Orleans. URL: *https://www.nationalww2museum.org/war/articles/scientific-and-technological-advances-world-war-ii#:~:text=Article%20The%20Scientific%20and%20Technological,day%20technology%20possible*.
[15] Bullard, E., Lovell, B., & Deacon, G. (1975). The Effect of World War II on the Development of Knowledge in the Physical Sciences [and Discussion]. Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences, 342(1631), 519–536. URL: *http://www.jstor.org/stable/7875*.
[16] Katzir, S. "In war or in peace:" The technological promise of science following the First World War. Centaurus. 2017; 59: 223–237. URL: *https://doi.org/10.1111/1600-0498.12161*.
[17] Oreskes, Naomi, and John Krige (eds), Science and Technology in the Global Cold War (Cambridge, MA, 2014; online edn, MIT Press Scholarship Online, 21 May 2015. URL: *https://doi.org/10.7551/mitpress/9780262027953.001.0001*, accessed 14 Oct. 2023.
[18] The official website of CARA (the Council for At-Risk Academics#). URL: *https://www.cara.ngo/who-we-are/our-history*.
[19] The official website of the global network Scholars at Risk. URL: *https://www.scholarsatrisk.org/get-help/*.
[20] Pierre Azoulay, Wesley H. Greenblatt, Misty L. Heggeness, Long-term effects from early exposure to research: Evidence from the NIH "Yellow Berets". Research Policy, Volume 50, Issue 9, 2021. URL: *https://www.sciencedirect.com/science/article/abs/pii/S0048733321001311*.

Risk[19]. Various countries offer scholarships to support prominent scholars at risk. For example, the participants of the National Institutes of Health's "Yellow Berets" program for medical schools graduates in the USA during the war in Vietnam were more likely to conduct research instead of limiting themselves to clinical practice.

During the Yugoslavian (Balkan) wars of 1991-1999 the **decrease in researchers' productivity** was connected not only to their emigration but also to disruption **of scientific connections and isolation from the international community[21]**. The post-war research on the renewal of producing scientific results[22] in 1999 and 2000 indicates an increase in publishing activity in Croatia (37.3% and 12.5%), Slovenia (10.9% and 52.8%), Macedonia (5% and 6%), and Montenegro (75% and 66%), while in Serbia, this index was still decreasing by 10.2% and 27.9% accordingly compared to 1998. The scientific production of Bosnia and Herzegovina decreased by more than 60%. Researchers attribute the decline to the fact that these regions experienced the most active combat[23,24]. A small number of lecturers and researchers in the field of biomedical sciences in Split and Osijek have published more publications per person in the period of war than most of their colleagues in two other centers[25].

In times of armed conflicts, scientists face **violations against fundamental human rights[26]** and **limitations of academic freedom** that make it impossible to conduct research. The desire to protect fundamental human rights and search for an ability to conduct research during armed conflicts may lead to collaborations with the warring parties that the other party will see as a collaboration with the enemy.

Armed conflicts and political misunderstandings can result in decisions **restricting scientists' freedom of movement**. For example, in Algeria, the Ministry of Higher Education and Scientific Research, reacting to "anti-Algerian" articles in a Moroccan magazine and tensions between the two nations, has prohibited the scientific community members to participate in conferences or seminars in Morocco. The Israeli government has implemented a directive restricting the travel of foreign scientists and students to the West Bank, along with limitations on their ability to engage in work within that region[27]. Freedom of movement is fundamental for high-quality scientific research as it contributes to exchanging new ideas and strengthens international academic collaboration to solve world problems. Democratic societies demand free movement for scientists, students, and their ideas. Restrictions on academic trips, whether pertaining to travel between countries, within a country, bans on entry or exit, or policies constraining the movement of scientists, pose significant obstacles to the international exchange of ideas—a hallmark of contemporary global research.

There are also known cases when a group of researchers at risk have come together, providing mutual support and creating

[21] Igić, R. The influence of the civil war in Yugoslavia on publishing in peer-reviewed journals. Scientometrics. 53, 447–452 (2002). URL: *https://doi.org/10.1023/A:1014833315145*.
[22] Ivanović, D., Jovanović, M., & Fritsche, F. (n.d.). Scientific Cooperation in the Republics of Former Yugoslavia Before, During and After the Yugoslav Wars. URL: *https://publica-rest.fraunhofer.de/server/api/core/bitstreams/7197b99f-2b20-4e36-8712-d8352e363a43/content*.
[23] Ivanović, D., Jovanović, M. & Fritsche, F. Analysis of scientific productivity and cooperation in the republics of former Yugoslavia before, during and after the Yugoslav wars. Scientometrics. 107, 499–519 (2016). URL: *https://doi.org/10.1007/s11192-016-1853-1*.
[24] Jovanović, Miloš & John, Marcus & Reschke, Stefan. (2010). Effects of civil war: Scientific cooperation in the republics of the former Yugoslavia and the province of Kosovo. Scientometrics. 82. 627–645. doi: 10.1007/s11192-010-0176-x.
[25] Lukenda, J. Influence of the 1991-1995 war on Croatian publications in the MEDLINE database. Scientometrics. 69, 21–36 (2006). URL: *https://doi.org/10.1007/s11192-006-0136-7*.
[26] For more information, visit: *https://www.un.org/en/about-us/universal-declaration-of-human-rights*.
[27] Ink, S. (2023, February 27). Free to think 2022. Scholars at Risk. URL: *https://www.scholarsatrisk.org/resources/free-to-think-2022/*.

opportunities to conduct research in a secure environment. A notable example of such collaborative efforts is the Ukrainian Free University[28] in Munich, Germany, founded by the Ukrainian scientific emigrants in 1921. Responding to the contemporary challenge posed by the full-scale Russian invasion, the National University of the Kyiv Mohyla Academy has initiated a proactive response by establishing campuses in Europe and North America[29].

The modern world is highly vulnerable, and a military conflict may arise in any country, so every country should develop its own policies for preserving human resources in science as a strategically important capital for renewal and development.

During an armed conflict, scientists, like all civilians, face violations against fundamental human rights within the conflict zone. Limitations on academic freedom are a risk specific to researchers.

Armed conflicts result in the loss of human capital for science through physical casualties, emigration, and the interruption of research. Next, we will analyze the impact of the full-scale Russian invasion on Ukrainian scientists and their ability to produce scientific results in more detail.

[28] The official website of the Ukrainian Free University: *https://www.ufu-muenchen.de/*.
[29] URL: *https://nv.ua/ukr/kyiv/kiyevo-mogilyanska-akademiya-vidkrivaye-kampusi-v-pivnichniy-americi-ta-yevropi-novini-kiyeva-50249924.html*.

Chapter II. Losses in Scientific Human Resources for Ukrainian Science During the War with Russia

2.1. Classification of Losses of Scientists During the War

In Ukraine, challenges related to human resources in science were present even before the full-scale invasion, as revealed by the analysis of the data from the State Statistics Service of Ukraine[30] for 2020. (Fig. 1, 2).

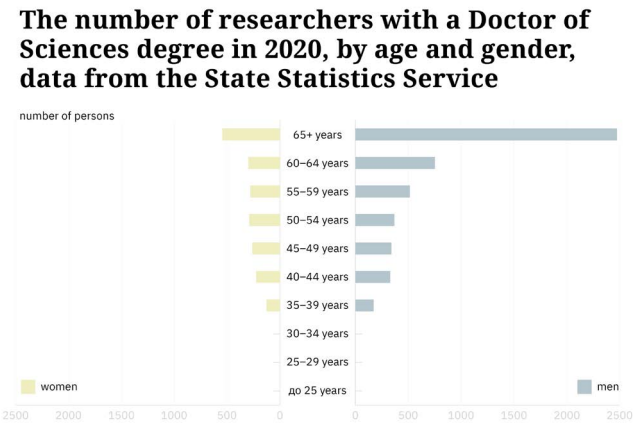


Fig. 1. The number of researchers with a Doctor of Sciences degree in 2020, by age and gender, data from the State Statistics Service, number of persons

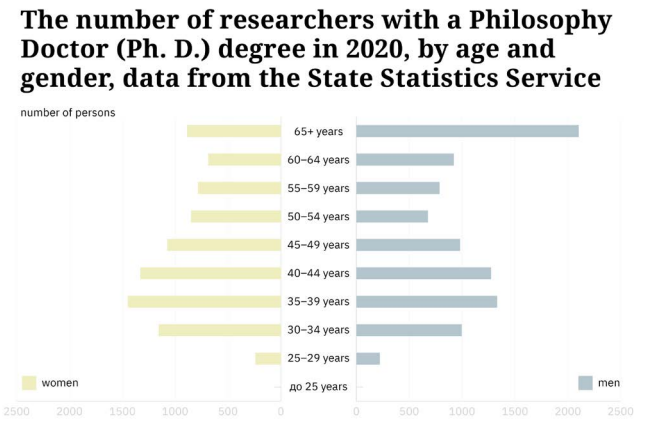


Fig. 2. The number of researchers with a Philosophy Doctor (Ph. D.) degree in 2020, by age and gender, data from the State Statistics Service, number of persons

The data analysis from the State Statistics Service of Ukraine underscores the problem of a **low influx of young personnel into research jobs already present in 2020**. Russia's full-scale invasion has made this problem even

[30] Scientific and innovative activity in Ukraine - 2020. URL: *https://ukrstat.gov.ua/*.

more severe, adding to the loss of scientists. At the same time, it is noteworthy to highlight a rise in the number of middle-aged female researchers (30-50 years), as during the armed conflicts, women and girls are the most vulnerable categories[31].

The deficit of young scientific personnel is a typical problem, in particular, for NASU. As of the beginning of 2023, the NASU institutions reported that 44.7% of employees holding scientific degrees had reached retirement age, amounting to 3976 individuals (compared to 42.7% last year). Among them are 1733 Doctors of Sciences and 2243 PhD, or, respectively, 71.6% of the overall number of DSc and 34.6% of the overall number of PhD who work in the NASU institutions. The overall number of young researchers decreased by 15.1% compared to 2021, and the number of young Ph. Ds decreased by 17.8%. In comparison to 2010, when the biggest number of young scientists in the NASU for the whole period of Ukrainian independence was documented, the overall number of young scientists decreased by 44%, and PhD decreased by 60%[32].

Due to the full-scale invasion, significant human capital in scientific sphere was lost because of the physical deaths of scientists, a temporary halt in research activities due to service in the Armed Forces of Ukraine (AFU), and the inability to continue research because of forced relocations, encompassing both internally displaced persons and emigrants. According to the "Free to Think 2022" report by Scholars at Risk (SAR), there were over 391 documented assaults on higher education communities across 65 countries and territories between September 1, 2021, and August 31, 2022. It was armed conflicts and political upheavals that became the most dangerous threats in this period for the academic freedom and institutional autonomy of the universities, communities, and values they represent. In particular, the war in Ukraine has led to disastrous consequences for scientific and educational institutions and science overall. The loss of scientists during the war and emergencies has created a situation where conducting research becomes virtually impossible, leading to a detrimental impact on scientific progress and innovation implementation.

These losses can be measured by identifying:
1) The actual number of scientists killed or wounded directly due to the war.
2) The destruction of research groups caused by the "brain drain" and the actual number of scientists who, due to forced relocation abroad, can't conduct research or have changed affiliations and are now conducting research in foreign scientific institutions.
3) The loss of knowledge and research interruptions due to physical obstacles, such as prolonged air-raid threats, power outages, destruction of equipment and laboratories, and data losses.
4) A decrease in the number of scientific outcomes, including publications, patents, and registration of copyright certificates.
5) A decrease in research and development funding (budget sequesters for research).
6) Pauses/support in the international science projects:

- Limitations on foreign travel for male scientists.
- Creation of programs to support scientists funded by the EU and other countries.

7) Learning and education as crucial components of renewing human capital in science.

The overall number of employees who worked in NASU as of 01/01/2023 was 26,167 persons, including 13,883 scientific officers, a decrease from the previous year's count of 14,212 by 329 persons (2.3%). The

[31] Lessons in war 2015 – Scholars at risk. (n.d.). URL: *https://www.scholarsatrisk.org/wp-content/uploads/2016/05/Lessons_in_War_2015.pdf*.
[32] Report of the National Academy of Sciences of Ukraine 2022. Report on the activities of the National Academy of Sciences of Ukraine in 2022 / NAS of Ukraine. — Kyiv : Akademperiodyka, 2023. — 593 p.

net weight of scientists in the overall number of employees of scientific institutions decreased by 1.8% compared to 2022 and is now 53.1%. The number of employees with a scientific degree in the NASU institutions, as of 01/01/2023, was 2,422 DSc and 6,487 PhD Compared to 2022 (2,490 DSc and 6,628 PhD), for the first time in many years, the number of doctors of sciences decreased by 68, and PhD — by 141. The number of scientific officers without a scientific degree has also decreased to 4,974 compared to 5,129 in 2021[33]. The median age of the scientific officers was 55.7 years; for Doctors of Sciences, it was 65.8 years; for PhD, it was 52.2. In 2022, 12 NASU employees lost their lives in combat against the enemy or as a consequence of shelling civilian objects by the aggressor.

As of July 20, 2023, the number of scientific employees unable to work at their regular workplaces reached 2,202 individuals, constituting 16% of the factual number of NASU employees. Among them, 1,497 persons (11%) left Ukraine, while 705 (5%) are internally displaced persons within the country. Of them, 3.9% work remotely, another 0.7% are on unpaid vacation, 0.5% are forced to stop working but receive 67%, and 0.3% are on paid vacation. The number of NASU employees currently in military service in AFU and other military formations is 197 persons (1.4% of the overall number of scientists in the NASU)[34].

According to the National Research Foundation, during martial law, only 57 of 169 teams are ready to continue the research and their halted projects, 62 teams may continue work under certain circumstances, and 50 collectives won't be able to continue with their research at all at this stage. In the structure of NASU, the highest number of emigrants belonged to the field of physics, with nearly 500 individuals: 293 scientific officers from institutions of the Departments of Physics and Astronomy and 202 scientific officers from the Department of Nuclear Physics and Power Engineering. Conversely, the fields with the lowest emigration figures were the Department of Literature, Language, and Art Criticism (34 persons), the Department of Mechanics (38 persons), and the NASU Presidium institutions (34 persons).

In 2022, the Ministry of Education and Science (MES) conducted a survey among representatives of universities and research institutions, receiving responses from 304 participants. The aim was to assess the situation regarding scientists' ability to carry out their professional work. The survey gathered data from 97 universities, covering 56,825 employees, including scientists, scientist-pedagogues, and young scientists. The data also included information on their locations and the ability to continue research as of October 2022 (Fig. 3).

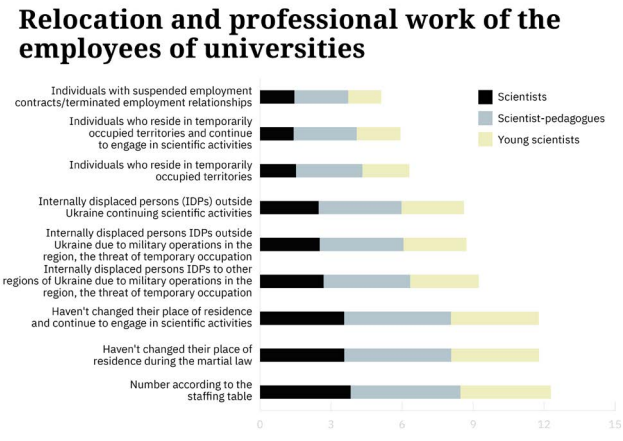


Fig. 3. Relocation and professional work of the employees of universities (sample of 97 universities)[35].

[33] The report on the activities of NASU in 2022. *https://files.nas.gov.ua/PublicMessages/Documents/0/2023/05/230509205447582-3171.pdf*
[34] The report on the activities of NASU in 2022. URL: *https://files.nas.gov.ua/PublicMessages/Documents/0/2023/05/230509205447582-3171.pdf*.
[35] The informative and analytic reference about the possibilities of doing their professional work by the subjects of scientific and scientific and technical work in universities and research institutions under martial law, including research by young scientists *https://mon.gov.ua/storage/app/media/nauka/informatsiyno-analitychni/2022/10/12/Inform-analit.dov.pro.mozhl.provadzh.subyekt.nauk.diyal.ZVO.12.10.2022.pdf*

Respectively, according to the results, in percentage: 73.6% of the employees haven't changed their location during the period of martial law; 72.4% haven't changed their location during the period of martial law and continue to conduct research, scientific and technological, scientific and organizational, and scientific and pedagogical work; 10.3% have relocated to other regions of Ukraine due to active combat in the region, the threat of temporary occupation, etc.; 10.2% have relocated to other regions of Ukraine due to active combat in the region or the threat of temporary occupation and continue their work; 7.6% have emigrated abroad; 7% have emigrated abroad and continue their work; 1.3% are on the temporarily occupied territories; 0.9% are on the temporarily occupied territories and continue their work; 0.4% are employees, with whom work contracts were put on hold or terminated.

Overall, it's sensible to divide the losses of the scientific potential into two groups: direct and indirect losses. Among the direct losses are forced migration, forced deportation of scientists from Ukraine to Russia, murder of civilians, and death while in service in AFU. Indirect losses encompass factors such as scientists not receiving their full wages, pauses in funding for grant programs, the inability to utilize intellectual and creative potential fully due to hostilities in Ukraine, insufficient investment in human capital due to the destruction of appropriate infrastructure, movement restrictions, etc. The list of indirect losses is not exhaustive and needs detailed processing and calculation in the post-war period. Overall losses can be assessed on three levels: losses within scientific communities, losses among individual scientists, and pauses in research within specific domains.

Thus, the losses of scientists caused by the war can be categorized into:

- physical losses of people, damage to well-being, and severe injuries;
- internal and external migration of specialists, relocation of entire collectives, dispersal of scientific communities;
- collaboration (conducting research in the interests of Russia), voluntary relocation to Russia, continuation of research within Russian territories;
- recruitment of researchers in service in AFU;
- quitting the profession or taking a pause in research;
- change of affiliation to scientific institutions of other countries.

The facts of physical losses of scientists: Oleh Amosov (economist), Yulia Zdanovska (mathematician), Vasyl Kladko (physicist), Andrii Kravchenko (chemist), Oleksandr Korsun (chemist), Oleh Andriitsev (historian), Yurii Kovalenko (archaeologist), Volodymyr Fedorov (biophysicist), Bizhan Sharapov (biologist), Yevhenii Osievskyi (anthropologist), Bohdan Sliushchynskyi (sociologist), Svitlana Kormiltsyna (senior lecturer of the chair of foreign philology of the Chernihiv Polytechnic National University), Natalia Toria (employee of the directorate of the Educational and Research Institute of Architecture, Design, and Geodesy of the Chernihiv Polytechnic National University)[36], and many others.

Collaborators, voluntary relocation to Russia with developments and research results: some Kherson State Agrarian University scientists[37].

Involvement of scientists in AFU: there are over 60 researchers and pedagogues of universities of the Rivne region currently in the Armed Forces of Ukraine and Forces of Territorial Defense: 34 people from the National University of Water and Environmental Engineering, 9 representatives of the National University "Ostroh Academy", 5 — from the Rivne State Humanitarian University,

[36] The rocket bombing of the center of Chernihiv: the names of all the seven dead became known https://suspilne.media/554717-raketnij-obstril-centru-cernigova-stali-vidomi-imena-vsih-semi-zagiblih/

[37] A family of researchers in Kherson became collaborators and transferred to work in the "Russian university" https://pivdenukraine.com.ua/2022/08/11/rodina-naukovciv-u-xersoni-stala-kolaborantami-perejshovshi-pracyuvati-v-rosijskij-universitet/

4 — from the Academician Stepan Demianchuk International University of Economics and Humanities, 7 from the Rivne Medical Academy, 2 — from Dubno Affiliation of the "Ukraine" University[38]. 14 polar explorers from the National Antarctic Scientific Center are now in AFU[39]. This list, as of the moment when research is underway, can't be complete for objective reasons; it only illustrates the active engagement of scientists in the AFU service.

Despite the relatively **broad classification of losses, their calculation turns out to be a significant problem** due to the war and the lack of consistent mechanisms for accounting within different entities such as the Ministry of Education and Science (MES), National Academy of Sciences of Ukraine (NASU), research institutions, and universities under the jurisdiction of other ministries, etc. There are separate public initiatives to collect data on the Ukrainian scientists who died from Russian aggression; as of November 7, 2023, there are 70 names on the list[40].

As Zoia Petrenko (the leader of the expert group on issues of science development, science directorate, and innovations in MES) and Oleksandr Stryzhak (deputy director of scientific work of the National Center "Junior Academy of Sciences of Ukraine") have noted in their interviews, before 02/24/2022, there was no accounting of human losses and relocations of scientists due to the Russian aggression.

MFA raised this question as a reaction to UNESCO's willingness to fund the preparation of a report on the damage inflicted on the scientific sector in Ukraine by the Russian war. Following the signing of a contract between the Center of the II Category of UNESCO and the Junior Academy of Sciences of Ukraine (JAS), the JAS initiated the "Damage from Russian Aggression: NAS of Ukraine" platform. This platform aimed to systematize data, including information on the relocations of scientists, detailing the number of scientists who left Ukraine due to the full-scale Russian aggression, categorized by country and the Ukrainian institutions they represented. Additionally, it accounted for the number of scientists who became internally displaced persons. However, the initial accounting primarily focused on scientists within the NASU structure.

To enhance the assessment process, a joint order from the NAS of Ukraine and MES of Ukraine on April 27, 2023, established an interdepartmental workgroup dedicated to preparing a comprehensive report on the damage inflicted on the Ukrainian scientific sector by Russian aggression. The interdependent workgroup involved the participation of representatives from the "Science at Risk'' project of the Kunsht non-profit. During the first meeting on July 5, 2023, besides the tables "Information on damaged and destroyed equipment of universities/research institutions due to the Russian military aggression" and "Information of the damaged and destroyed real estate of universities/research institutions due to the Russian military aggression", the questionnaire "Losses of the scientific human resources of universities/research institutions due to the Russian military aggression" and a table "Information on the state of work organization of the researchers and pedagogues of universities/research institutions due to the Russian military aggression" was submitted for discussion. As a result of this meeting, these questionnaires were sent to institutions for completion, although many institutions are yet to submit their responses. Nonetheless, these questionnaires are currently key instruments to document human losses, relocations, and changes in researchers' statuses.

[38] Over 60 scientists from Rivne region are defending Ukraine in AFU. URL: *https://armyinform.com.ua/2022/05/23/ponad-60-naukovcziv-iz-rivnenshhyny-boronyat-ukrayinu-v-lavah-zsu/*.
[39] URL: *https://www.ukrinform.ua/rubric-ato/3681492-u-lavah-zsu-vouut-14-polarnikiv.html*
[40] War and science: Ukrainian losses. URL: *https://my.science.ua/vijna-ta-nauka-ukrayinski-vtraty/?fbclid=IwAR2OcMf8CaZAhN3WnGa1Traqaer8KZ-_RbWoH9Jd4_4vHkLXrYjvTRbPuAk*.

Despite the absence of a unified accounting system, MES continues to provide isolated data on losses in the scientific sphere. As of the beginning of 2023, 91 research institutes and universities had suffered from Russian shelling, with four of them completely destroyed. The Donetsk and Luhansk regions, situated in active warfare zones, experienced a significant disruption of scientific and educational activities. Many scientists were forced to evacuate to other regions of Ukraine or move abroad to save their lives from the war: over 6000 scientists are currently abroad. Survey results from the UAScience. Reload project reinforce this situation, highlighting that the material situation has worsened for 83.8% of scientists. Moreover, 47.2% of scientists stayed in Ukraine without relocating due to the war, 38.1% are internally displaced persons, and 14.7% are abroad. The war has led to the loss of work, a portion of wages, and funding for projects among Ukrainian scientists[41].

2.2. Results of Analytical Research and Sociological Surveys

The impact of the war on Ukrainian scientists became a subject of research by many analytical groups; in particular, a separate special scientific publication, "Academic Management In War", was published[42]. In the survey "Ukrainian scientists in wartime"[43], held by the National Research Foundation of Ukraine in April 2022, the majority of 2,173 respondents (72.9%) answered "No" to a question about whether it's possible to conduct research in the same volume as before the war. The respondents have noted four main reasons for this inability: a sense of danger, the necessity to work in a location currently inaccessible, technical challenges like internet and communication issues, and apathy or lack of interest as responses to stress and trauma. The research[44] focusing on Ukrainians who were forced to relocate and have a temporary shelter in Germany and Poland showed that 4% in Poland and 6% in Germany were employed in education and research.

In the UAScience.Reload project, multiple waves of surveys were conducted. In the first wave (April 1 to May 2, 2022), 2,173 respondents were surveyed, and it was found that 73.7% of those who stayed in Ukraine at the time of the research were living in places without active combat (including 19.9% in the capital Kyiv), while 7.0% were in active warfare zones. Another 19.3% lived in cities and villages without active combat but were in the zone of warfare. The scientists'

[41] Scientists during the war: identifying needs and overcoming challenges *https://svit.kpi.ua/2022/06/27/%D0%BD%D0%B0%D1%83%D0%BA%D0%BE%D0%B2%D1%86%D1%96-%D0%BF%D1%96%D0%B4-%D1%87%D0%B0%D1%81-%D0%B2%D1%96%D0%B9%D0%BD%D0%B8-%D0%B2%D0%B8%D1%8F%D0%B2%D0%B8%D1%82%D0%B8-%D0%BF%D0%BE%D1%82%D1%80%D0%B5%D0%B1/*

[42] Business Perspectives - issue #2 (spec. issue) Special issue: Academic management in war. Special issue: Academic management in war. (2023).

[43] Ukrainian scientists and educators in wartime. American Scientist. (2023, February 17). *https://www.americanscientist.org/article/ukrainian-scientists-and-educators-in-wartime*

[44] A. Zymnin, M. Kowalski, M. Sielewicz, K. Klages, S. Kowcun, S. Bryzek, E. Dabrowska, D. Koszykowska, F. Stelmach. Report Ukrainian refugees vocational activation in Germany and in Poland. A sociological study.2022, p. 27

location played a crucial role in their perception of danger and relative safety and their ability to continue research. Since the beginning of the war, many scientists also underwent changes in their participation in scientific projects. Over the last three years, 59.0% of researchers did not participate in grant or competitive projects, while 24.9% were involved in projects financed by international donors, and 28.2% in projects funded by the Ukrainian government and/or national foundations. The ongoing warfare and instability posed challenges for scientists engaged in government-funded projects, with 29.1% reporting project delays or execution problems. The majority of respondents (83.8%) said that their financial conditions worsened compared to the pre-war conditions. 39.7% had the money for food only, and 27.7% could buy food and clothes but had trouble buying long-term-use goods.

The data analysis has shown that during Russia's full-scale invasion, the main problems for most scientists were the feeling of danger and financial instability, negatively impacting their ability to conduct research. For scientists residing in areas with active combat, the feeling of danger and technical issues were more frequently cited as obstacles to research compared to those living far from active hostilities. For scientists abroad, the primary reason for research interruptions was the inability to be physically present at their workplaces, a necessity due to the specific nature of their work.

Overall, the results of the first wave of the research show that Ukrainian scientists are facing hardships during the war. They face many challenges, like financial instability, inability to concentrate on research fully, and limited access to resources. However, they also demonstrate a high level of eagerness to assist in the work related to strengthening defense capabilities and reviving Ukraine.

The second research of the UAScience.Reload initiative was held online with the help of the LimeSurvey app from September 26 to November 24, 2022[45]. Overall, 1,729 scientists participated in the survey, creating a significant volume of data for further analysis. In their respective locations, 53.2% of the surveyed scientists stayed in Ukraine without changing their residence; another 34.8% were internally displaced with 52.8% returning to their pre-invasion places of residence. At the same time, 86.8% of respondents continued to receive wages in the educational institutions, while 5.7% did not receive wages from any institution. At the moment of the survey, 12% of scientists were abroad.

Like in the previous wave, among the major obstacles hindering full research engagement included the freezing of many research projects due to the war, technical limitations such as internet and communication network disruptions, power outages, and the absence of safety. However, the feeling of danger intensified, particularly related to heightened rocket attacks on Ukraine in October. Consequently, scientists abroad felt comparatively safer than their counterparts in Ukraine. As the shelling continued into November 2022, respondents assessed the safety of their places of residence as worse than in previous periods. Under these circumstances, continuing research became almost impossible, with many scientists experiencing heightened stress and apathy.

The third research initiated by UAScience.Reload was held from July 5 to August 13, 2023. The majority of respondents (90% of scientists) were in Ukraine at the moment of the survey, and the rest 10% were abroad. 51.8% of those who were abroad were planning to return to Ukraine with the end of the war, and another 20.6% planned to return in 1-3 years after the end of the war. 11.4% of the scientists don't plan to return to Ukraine. Overall, 50.3% of the scientists didn't

[45] Key Results of The Survey on The Needs of Ukrainian Scientists (Second Wave Report) *https://www.uascience-reload.org/2023/07/06/key-results-of-the-survey-on-the-needs-of-ukrainian-scientists-second-wave-report/*

change their place of residence because of the war.

Around 88% of the respondents reported continuing to receive wages from the institution(s) where they worked, while 11.2% did not receive wages from at least one institution. About 43.3% noted an increased workload due to staff reductions and the absence of some employees involved in the Armed Forces of Ukraine or Territorial Defense. Delays in wage payments were mentioned by 32.1% of scientists, and 29.3% did not work full-time, with 27.4% indicating they did not receive their wages in full.

54.3% of the scientists noted that they couldn't do as much research as before the war (usually those who were both researchers and pedagogues). The main reasons for the inability to conduct research in full were still the absence of feeling of safety and technical reasons (problems with internet connection and communication networks, power outages, etc.). Those who were abroad highlighted the very fact of being in a foreign country as an obstacle to conducting research.

Compared to the previous surveys, the dynamics in the direction of self-organization of scientists in grant activities and participation in international projects became more pronounced. Scientists exhibited greater participation in projects funded by the EU/ international donors (26.1%) compared to those funded by the Ukrainian government/ national foundations (18.3%) and businesses (5.7%), as the letter two tend to be put on hold more often because of the war. They also started to show more initiative in informing about their scientific requests and needs and getting assistance from international foundations, foreign universities, scientific institutions, etc.

Similar to these waves of surveys by UAScience.Reload, there was also a research conducted by the ScienceForUkraine[46] initiative, which provided insights into the social and demographic status of Ukrainian researchers abroad, their professional preferences, and the challenges they are facing (619 respondents from 35 countries — employees of Ukrainian research institutions as of 02/24/2022 either staying or relocating on the day of the invasion or later). At the moment of the survey, most of the respondents continued to work remotely in their institutions on a regular basis (30.1%), and almost half of the rest were on unpaid vacation (28.7%). Subsequently, conditions for the respondents changed slightly, with 14.8% continuing to work remotely but under different conditions, and a similar number being on unpaid vacation or no longer working (17% overall). A smaller percentage (9%) were the scientists who were forced to officially look for a job in a foreign institution, either through participating in academic mobility, scientific internships (3.2%), or quitting their institutions (1.9%). Almost 0.5% found a non-academic job abroad, and 7.2% were in social assistance programs. Isolated cases included the continuation of postgraduate education or doctoral studies, forced unpaid academic vacation, maternity leave, and the necessity to visit Ukraine occasionally, etc. The survey results have shown that the researchers' current professional situation in their Ukrainian institutions didn't influence their actions, as almost 85% were looking for some kind of assistance outside of Ukraine. Over half received income from wages or scholarships from their institution (though not in full), and two-thirds (67.7%) received full payments. Overall, over 20% of respondents remained unemployed, and only 4.1% were not actively seeking a job.

On the scale of the importance of access to assistance, homes and wages were at the top;

[46] Maryl, Maciej, Jaroszewicz, Marta, Degtyarova, Iryna, Polishchuk, Yevheniia, Pachocka, Marta, & Wnuk, Magdalena. (2022). Beyond Resilience: Professional Challenges, Preferences, and Plans of Ukrainian Researchers Abroad. Zenodo. https://doi.org/10.5281/zenodo.7380509. URL: https://zenodo.org/record/7380509

the next was family assistance, the ability to work remotely, and health insurance. The ability to work part-time, IT equipment, and psychological assistance were considered less important.
The results of the survey, initiated by the Board of Young Scientists of MES of Ukraine[47] in 2022, also show the needs of Ukrainian scientists in dangerous circumstances. The survey was held among 303 representatives of universities and research laboratories who stayed in Ukraine and were in different levels of danger. Among the respondents, 74% were researchers and pedagogues, 12% were postgraduates in universities and research institutions, and 11% were scientific officers. While a small number of respondents emphasized the importance of preserving their homes and financial aid, the majority identified the ability to work remotely as a critical necessity. The survey revealed specific needs, including free access to databases, libraries, and archives (such as Statista, WoS, SCOPUS, Reaxys, SciFinder, Bloomberg, JSTOR, etc.), access to research equipment and laboratories abroad (enabling the analysis of specimens in foreign laboratories), licensed software, cloud work environments, and the establishment of international connections to facilitate joint grant applications with foreign colleagues. Remote work was seen as an efficient way for scientists to continue their daily work and ensure the continuity of their research. Ideas for remote collaboration encompassed joint research projects (including internships), virtual mobility, remote lectures in foreign universities, collaborative publications of articles and monographs, development of programs for academic mobility after the war, remote data array processing using the equipment of foreign universities, and collaboration in programs for the economic revival of Ukraine, among other initiatives.

Given the high level of support for similar types of activities mentioned by the respondents, we may hypothesize that, at the moment of the research, information about available opportunities and offers might not have been reaching its intended audience in a structured manner, or obtaining such information could have been challenging due to various circumstances. Therefore, the need for communication measures is especially relevant to overcome all obstacles in accessing and actively participating in all the offers.

Thus, the emergence of various kinds of surveys and research on the needs of scientists at risk can be attributed primarily to **the absence of reliable and full information on the dynamics of human losses in the scientific sphere, the relocation of scientists, and their most acute needs in the extreme conditions of the war in Ukraine. Neither the government nor scientists themselves had clear data on the issue of preserving human resources in science**. It turned out to be problematic to do a unified accounting of the losses of scientists and scientific schools in the country during the war and also find out the reasons for losses (physical destruction, change of affiliation, collaboration). Instead, we discovered that every research institute or university has such information.

Due to the full-scale invasion of Russia in Ukraine and the forced relocation of people to safer places, distinct groups of scientists have emerged:

- Scientists who stay in Ukraine, who may be on the occupied territories, in the zone of active combat, or a relatively safe territory;
- Scientists in service in the AFU;
- Scientists who found temporary shelter abroad;
- Scientists demobilized from AFU.

Considering the ability to conduct research, more than half of the researchers on Ukrainian territory or abroad claim they are able to continue research. However, scientists actively

[47] Polishchuk Y., Moskvina V., Degtryarova I., Galat M., Makaruk L. (2022), Assessing the needs of Ukrainian scholars in danger circumstances. The study of the Council of Young Scientists under the Ministry of Edication and Science of Ukraine, Kyiv, 4 April 2022. *https://uni-freiburg.de/universitaet/wp-content/uploads/sites/3/2022/05/Remote-work-UA-scholars-at-risk.pdf*
[48] One Year of War in Ukraine: Exploring the impact on the science sector and supporting initiatives. International Science Council. (2023, September 20). *https://council.science/publications/ukraine-conference-report-2/*

serving in the Armed Forces of Ukraine (AFU) find their professional scientific lives unavoidably paused[48]. The scientists demobilized from AFU are returning to scientific work after a significant break and in a difficult mental state. Female scientists with children who found temporary shelter abroad point to the difficulties of combining research with giving proper care to their children. Scientists and their ideas are important for sustainable economic growth, and university research catalyzes local innovations and employment. The loss of scientists from Ukraine and damage to the research system can lead to long-term consequences to Ukrainian science and hinder the post-war economic revival of the country.

Thus, to preserve scientists during the war, it's important to 1) eliminate threats to their physical destruction, which, among other things, can be provided by first aid skills; 2) keep people in the profession, which means developing mechanisms to support scientists' ability to conduct research independently of their affiliation with specific groups mentioned earlier; 3) develop mechanism of succession and preservation of scientific schools in case of physical death of their scientific supervisor.

Chapter III. The Impact of War on Producing Research Results

The scientific results of researchers usually appear as publications, patents, and technological achievements. These results can be measured in quantitative indicators (for example, the number of received patents or published articles) or in qualitative impact indicators (for example, citation rates). Usually, war leads to a decrease in producing scientific results; in particular, in Ukraine, it has led to a 10% decrease in the number of scientific publications published by Ukrainian scientists primarily due to the loss of human resources and reduction in international collaborations[49]. In 2022, the number of applications filed for security documents based on scientific and technical works funded by the general state budget decreased by 33.9% (by 733 units) compared to 2021, amounting to 78.6% of the total number of applications filed for security documents. Also, compared to the previous year, the number of applications submitted abroad has decreased by 71.4%[50].

As of 08/15/2023, in the period from 2020 to 2023 , the international scientometric database Scopus has documented 73,040 publications of Ukrainian scientists and researchers (scientists that indicated affiliation with a Ukrainian institution are meant here), including articles published in collaboration with foreign researchers from 159 countries (Fig. 4). In quantity we have, publications per year: 2020 — 21,430, 2021 — 22,171, 2022 — 19,846, 2023 - 9,588.

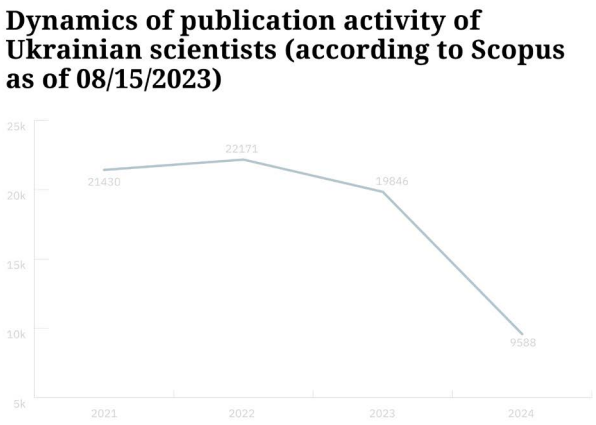


Fig. 4. Dynamics of publication activity of Ukrainian scientists (according to Scopus as of 08/15/2023)

If we compare the dynamics of the two stages 1) 2020-2021, and 2) 2022-2023, we have 43,601 publications and 29,434 publications respectively, so during the war, the number of scientific results has somewhat decreased (this may be the evidence of the fact that

[49] Ganguli, I., & Waldinger, F. (2023b, August 22). The impact of the war in Ukraine on science and Universities. CEPR. *https://cepr.org/voxeu/columns/impact-war-ukraine-science-and-universities*

[50] Research and scientific and technological work in Ukraine in 2022. (n.d.-c). *https://mon.gov.ua/storage/app/media/nauka/2023/07/25/Nauk-analit.dopov.Naukova.ta.nauk-tekhn.diyaln.v.Ukr.2022-25.07.2023.pdf*

scientific careers of many Ukrainians were broken because of the war, preventing them from continuing their work both at home and abroad), but this index isn't as critical, which proves the high mobilizational potential of scientists and their search for opportunities to continue scientific activities (such as grants, international support programs, foreign donors, etc.), development of inner academic mobility, etc.

By the field of knowledge, the top 5 of publication activity are: engineering, physics and astronomy, computer sciences, material science, and medicine. This distribution remains practically unchanged until 2023, but the decrease in publication activity in numbers (almost twice less) significantly impacted almost all fields of knowledge (Table 2). Also, in 2022, the number of publications where authors affiliate themselves with such institutions as NASU, Taras Shevchenko Kyiv National University, Lviv Polytechnic National University, National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute", V. N. Karazin Kharkiv National University, etc., has decreased by almost two times.

During 2021-2023, the dynamics of funding research in Ukraine have changed significantly: although the main subjects for funding are still MES, NASU, and NRFU, before the war, in the list of organizations sponsoring research, a large part consisted of the Russian Foundation of Fundamental Research (in such fields as physics and astronomy, engineering, material science, Earth and planetary sciences), Russian Scientific Foundation (physics and astronomy, material science, chemistry, engineering, biochemistry, genetics, and molecular biology), Ministry of Education and Science of Russia (chemical engineering, material sciences, chemistry, engineering), Russian Academy of Sciences (physics and astronomy, engineering, agriculture and biology, material science), etc. This means that Ukrainian scientists who continued to affiliate themselves with Ukrainian institutions could actively participate in research funded by Russiaor worked concurrently in research institutes of Russia (or vice versa — Russian scientists entered scientific topics of Ukrainian institutions). Almost every year, organizations from Russia were among the top 5 sponsors of scientific research in various capacities. With the beginning of Russia's full-scale aggression against Ukraine in 2022, the picture began to shift towards the exclusionRussia from funding organizations; instead, collaboration of Ukrainian scientists with institutes and organizations from Europe, the USA, China, and the UK has been strengthening. While in 2020-2021, 762 publications were financed by Russia, in 2022-2023 this number fell to 264 publications, with only around 60 publications in 2023, mainly in physics and astronomy, chemistry, engineering, material sciences, computer sciences, etc. The full dynamics of financing research in Ukraine from Russian sources look like this (Fig. 5):

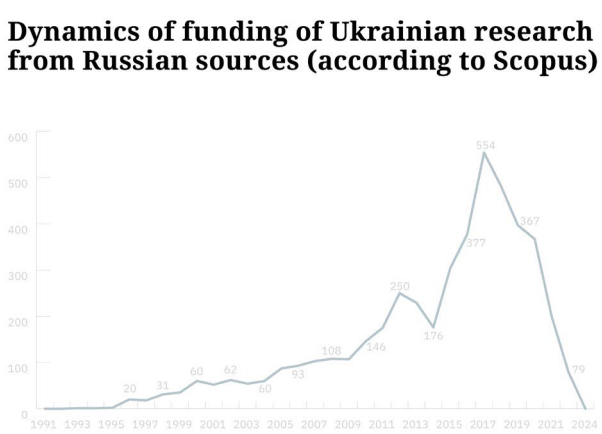


Fig. 5. Dynamics of funding of Ukrainian research from Russian sources (according to Scopus)

A similar situation is observed in the collaboration of Ukrainian scientists with foreign researchers. The top 10 countries with whom there was research collaboration were Poland, Germany, the USA, Great Britain, China, France, Italy, the Czech Republic, Greece, Slovakia, and Russia.

Before the full-scale war, Russia was second on this list (2020-2021), and starting with 2022, it moved to the 4th position, and from 2023, it's in the 11th position — out of nearly 10 thousand publications, only 258 were written in collaboration with Russia (fields: physics and astronomy, medicine, material science, engineering, chemistry, agriculture, and biology). Even though Ukrainian and Russian scientists continue to co-author much foreign research, there's a clear tendency towards narrowing the spheres of such collaboration and minimize intersection points of the scientific interests of researchers.

The contextual peculiarities of Russia's rootedness in Ukrainian science continue to negatively influence its further development and post-war recovery . After all, it's also a matter of international image, especially in terms of building close connections with foreign partners and donors. Ukraine and Russia have a relatively long shared historical context, and, respectively, it's hard to reformat the lasting connections in the research sphere in a relatively short period. More active collaboration with Western partners and EU countries contributes to distancing from this tradition, but at the level of international scientific projects and international consortia, the issue of collaboration remains relevant. The Law of Ukraine #2299-IX from 06/19/2022, "On the Termination of the Agreement Between the Government of Ukraine and the Government of the Russian Federation About Scientific and Technological Collaboration[51]" set a course to terminate collaboration of such kind with the aggressor state. This limitation, at the very least, aims to halt the strengthening of Russian science at the expense of the work of Ukrainian scientists. For example, the National Research Foundation of Ukraine amended its ethical principles, according to which scientists cooperating with Russian citizens and institutions on the occupied territories lose the possibility of getting funding[52]. Overall, the unwinding of scientific ties, scientific and technological cooperation, and joint projects with Russia has been gradually happening for almost two years already, and not only in Ukraine.

As an exception, it's taken into account that the issue of cooperation with Russia doesn't relate to instances of participation of Ukrainians in international scientific projects, consortia, and simply large collectives of authors formed mostly by scientists from developed countries, as the exclusion of Ukrainian participants from such collectives because of the presence of Russians in them will only reduce the representation of Ukrainian science in the world. Past cases of cooperation that are impossible to cancel or revise are also considered.

The scientometric analysis of Ukrainian scientific collaborations in 2019-2022, conducted by Lithuanian researchers[53], is also demonstrative in this context. They have analyzed authorship structure in 2,365 publications in open-access magazines of MDPI[54] and concluded that the full-scale war significantly reduced the network of cooperation of our scientists: in 2019, they collaborated (to varying extents) with various European countries, but in 2022, the main scientific partner was Poland (Fig. 6).

[51] Ganguli, I., & Waldinger, F. (2023b, August 22). The impact of the war in Ukraine on science and Universities. CEPR. *https://cepr.org/voxeu/columns/impact-war-ukraine-science-and-universities*.

[52] Provision "About the Adherence to Principles of Scientific Ethics and Prevention of the Conflict of Interests During Holding Competitive Selections, Scientific and Technological Expertise of Projects, and Consideration of Applications for Getting Grants Submitted to the National Research Foundation of Ukraine". URL: *https://mon.gov.ua/storage/app/media/nauka/2023/07/25/Nauk-analit.dopov.Naukova.ta.nauk-tekhn.diyaln.v.Ukr.2022-25.07.2023.pdf*.

[53] Robertas Damaševičius, Ligita Zailskaitė-Jakšt The Impact of a National Crisis on Research Collaborations: A Scientometric Analysis of Ukrainian Authors 2019–2022. *https://doi.org/10.3390/publications11030042 https://www.mdpi.com/2304-6775/11/3/42*

[54] *https://www.mdpi.com/*

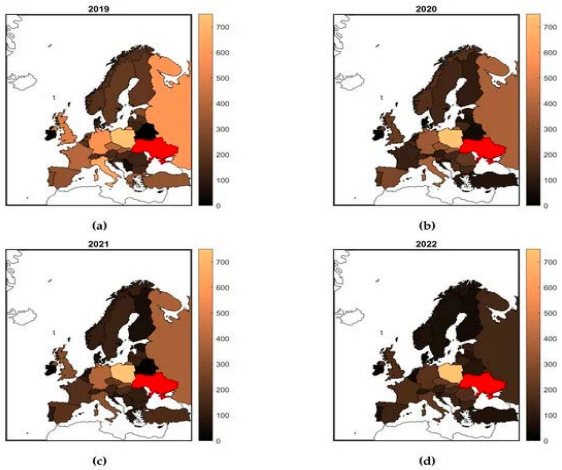


Fig. 6. International collaboration between researchers from Ukraine and other countries of Europe in 2019-2022. A lighter color means broader international collaboration. Ukraine is marked with red color[55].

The map illustrates the geographical distribution of European partnerships of authors with Ukrainian affiliations in 2019-2022 and demonstrates the decrease in the number of international collaborations with researchers from other countries (except Poland) in 2022 compared to the period of 2019-2021. At the same time, the venues for publications haven't changed: in 2019, the most popular place for publications was the "Energies" magazine; in 2020, it was overtaken by the magazine "Applied Sciences and Materials"; in 2021 and 2022, "Materials" and "Energies" remained the two leading magazines, which is the evidence of the stability of research topics of Ukrainian scientists. Most publications of Ukrainian researchers in open-access magazines relate to health sciences, medical and natural sciences, which is evidence of the greater resilience of these fields to the impact of crisis factors. Thus, according to this research, the dynamics of the Ukrainian scientists' research network has demonstrated a decrease in the variety of international collaborations in 2022. It should be noted that these are only data from MDPI (the sample is significantly limited to the open-access magazines). Still, at the same time, the quantitative indices show that the scientific career of many Ukrainians was terminated or put on hold due to war, and Ukrainian science has gone through serious changes, and therefore some academic structures have to adapt to survive in the reality of the war.

[55] The Impact of a National Crisis on Research Collaborations: A Scientometric Analysis of Ukrainian Authors 2019–2022. *https://doi.org/10.3390/publications11030042 https://www.mdpi.com/2304-6775/11/3/42*

Chapter IV. Research Practices of Ukrainian Scientists During the War

4.1. International Support for Research of Ukrainian Scientists

Despite Russia's full-scale invasion, the destruction of Ukrainian research infrastructure and universities, the brain drain, and the decrease in funding, Ukrainian scientists are making concerted efforts to ensure the continuity of research and the educational process. In particular, educators and scientists can join internal and international exchange programs thanks to the collaboration with partners. Paradoxically, this difficult period has served as a catalyst for the development of Ukrainian science as, from the very beginning of the full-scale war, it has received significant moral and financial support from various national scientific communities and international scientific organizations. The initial response from the global scientific community focused on providing opportunities, primarily for refugees leaving the country.

Before the beginning of the full-scale invasion (02/04/2022), the Ministry of Education and Science (MES) had published recommendations on emergency and war response[56] on its website. However, not all of these recommendations were universally applicable across regions with varying circumstances. Immediately after the beginning of the invasion on February 24, 2022, scientists-volunteers concerned about the adverse effects on their colleagues' work, initiated numerous **online resources**. They were mostly open electronic tables with a list of short-term and long-term vacancies, internships, and other types of

[56] *https://mon.gov.ua/storage/app/media/civilniy-zahist/2022/03.02/Broshura-U.razi.nadzvychaynoyi.sytuatsiyi.abo.03.02.pdf*

support[57]. Some of these resources appeared spontaneously from concerned individuals and groups, while others became a part of larger-scale support initiatives for Ukrainian scientists and educators (#ScienceForUkraine, Council for At-Risk Academics (CARA), ERA4Ukraine, European Federation of Academies of Sciences and Humanities (ALLEA), ERC4Ukraine, The Guild, IIE Scholar Rescue Fund, Labs Supporting Ukrainian, Scientists, LeCollègede France, OEG Connect, Philipp Schwartz Initiative of the Humboldt Foundation, MSCA4Ukraine, Saving Ukrainian Cultural Heritage Online (SUCHO), Scholars at Risk (SAR), Scientists and Engineers in Exile or Displaced (PAS/NAS SEED) program, Shevchenko Emergency Fund, Swiss National Science Foundation, The Association for Slavic, East European & Eurasian Studies, The UK-Ukraine Twinning Initiative, Ukrainian Global University Initiative, Ukrainian Scholar Placement Database). There were also calls to help those who were studying abroad already but couldn't continue their programs[58].

In order to aid the Ukrainian academic community during the war and sustain Ukrainian science's presence globally, a group of scientists-volunteers from all over the world initiated #ScienceForUkraine (SfU)[59] (it's already registered as an NGO). They organized a virtual international network and a central database of international opportunities available to Ukrainian researchers. SfU serves as an internet portal, providing Ukrainian scientists with information on international support programs, grants, scholarships, vacancies in foreign institutions, etc. and direct contacts with those willing to assist in preserving their scientific engagement. Besides education and work, some research institutions also offer short-term scholarships, free meals and residences, as well as aid to the families of refugees. SfU has helped around a thousand scientists find vacancies outside Ukraine, and around ten new possibilities appear every week. Initially SfU primarily focused its efforts on helping scientists who left Ukraine, now it tries to help researchers who stayed in the country as well (for example, they started 70 scholarships together with a global initiative, "Economists for Ukraine", 2022).

A special group from the University of Oregon has posted a list of laboratories ready to support relocated Ukrainian scientists. This initiative gained rapid popularity, and within 40 days of its creation, the list contained over 2000 vacancies in Europe and the USA, showcasing the widespread willingness of laboratories to support their Ukrainian counterparts (Labs Supporting Ukrainian Scientists). The Sam Houston University (Ukrainian Scholar Placement Database) also created a similar list. The Ukrainian Scholar Placement Database developed by Sam Houston University contains a list of funding programs available for researchers from Ukraine. At the same time, other communities, like the Saving Ukrainian Cultural Heritage Online (SUCHO), focused on activities like web archiving, digitization, and preserving the heritage of Ukrainian cultural institutions.

The networks of collaborations that existed before the war were also used to launch or strengthen support measures quickly. The extant pre-war structures, including **national societies and academies**, initiated a group of initiatives[60]. For example:

- The National Academy of Sciences (NAS), USA, and the Polish Academy of Sciences (PAS) launched a program supporting research visits to the Institutes of PAS[61];
- The Office of Science of the Department

[57] Maryl M, Ivashchenko Ov, Reinfelds M et al. Addressing the needs of Ukrainian scholars at risk. Nat Hum Behav. 2022;6:746–7. 10.1038/s41562-022-01387-7
[58] Heyman A. OU's Ukrainian community asks the administration for financial assistance. Oakland Post. 2022. *https://oaklandpostonline.com/41529/campus/ous-ukrainian-community-ask-administration-for-financial-assistance/*. Accessed 1 March 2023.
[59] *https://scienceforukraine.eu/listings*
[60] Scientists without borders: lessons from Ukraine. GigaScience, Volume 12, 2023, giad066. URL: *https://doi.org/10.1093/gigascience/giad066*.
[61] URL: *https://pan.pl/en/long-term-program-to-support-ukrainian-research-teams/*.

of Energy of the USA provides funding for payments for Ukrainian scientists. While the funding is directed to scientific institutions in the USA, it encourages the employment of Ukrainian students, professors, and researchers. The Department covers expenses related to employment, moving to the USA, and assistance in obtaining visas[62];
- The American Chemical Society has joined the National Academy of Sciences of the USA to support researchers who were forced to leave Ukraine[63];
- The program NAS Scientists and Engineers in Exile or Displaced (SEED) helps scientists move and continue their work in Poland and other neighboring countries[64];
- The German Service of Academic Exchange (DAAD) has launched a website of the National Academic Contact Center for Ukraine[65];
- The Finnish Scientific Society opened a competition for grants for Ukrainian scientists in Finland[66];
- The Austrian Academy of Sciences offered an emergency call for Ukrainian researchers in their program of academic mobility "Joint Excellence in Science and Humanities" (JESH)[67];
- The French Association of female scientists Femmes&Sciences helps Ukrainian female scientists who have suffered from war and are or plan to be in France[68];
- The European Commission has opened a portal of European Research Area for Ukraine;
- The British Twinning Initiative, a model of collaboration between universities, includes British-Ukrainian research and innovation grants for twinnings from UKRI's Research England, to give fellow researchers an opportunity to continue their research and innovative collaboration[69];
- The program "Horizon Europe" offers free participation in the Program of Research and Education of Euratom that encompasses nuclear research and innovation, to support Ukrainian scientific community, etc.

Besides national academies, global **research networks and international associations** specializing in certain fields (Scholars at Risk; IIE Scholar Rescue Fund, Philipp Schwartz Initiative of the Humboldt Foundation in Germany, Council for At-Risk Academics in Great Britain, and others) were also actively engeged. With the support of their governments, philanthropists, and large corporate sponsors, they have quickly focused their efforts and created opportunities for relocated Ukrainian scientists. In particular, Scholars at Risk (SAR)[70] protects scientists who face serious threats to their lives, freedom, and well-being by organizing temporary research and professor vacancies in institutions within the network and providing consulting services. Since its foundation in 2000, SAR has assisted over 2000 researchers. In 2022-2023 alone, SAR received over 1300 applications and provided a wide range of direct services, including offering 189 vacancies to scientists at risk.

In a relatively short time, some universities, research institutes, and academic organizations from all over the world also started to offer scholarships, paid vacancies, and other types of support. These and other processes have given an impetus to creating new international partnerships and **upscaling the academic mobility programs**. During the first month of the full-scale invasion, these changes were chaotic, but currently, there is a growing emphasis on organizing and systematizing

[62] URL: *https://science.osti.gov/grants/FOAs/Open*.
[63] URL: *https://www.acs.org/pressroom/newsreleases/2022/april/acs-joins-nas-in-effort-to-support-researchers-forced-to-flee-ukraine.html*.
[64] URL: *https://www.daad-ukraine.org/en/*.
[65] URL: *https://nrfu.org.ua/news/finske-tovarystvo-nauk-ta-literatury-dopomogaye-postrazhdalym-vid-vijny-ukrayinskym-naukovczyam/*.
[66] URL: *https://nrfu.org.ua/news/avstrijska-akademiya-nauk-dopomogaye-postrazhdalym-vid-vijny-ukrayinskym-naukovczyam*.
[67] URL: *https://www.femmesetsciences.fr/news/accueil-des-femmes-ukrainiennes*.
[68] URL: *https://www.eua.eu/about/who-we-are.html*.
[69] URL: *https://www.ukri.org/councils/research-england/*.
[70] URL: *https://www.scholarsatrisk.org/*.

these efforts to provide more effective assistance.

The network of international academic mobility in Ukraine has become possible in many ways thanks to the EU's Erasmus+ program[71]. Currently, Ukrainian organizations (Ukraine is a country-partner in the Eastern Partnership — Region 2) have access to 35 competitions, taken care of by the National Erasmus+ Office in Ukraine (NEO) that, in tight collaboration with the Delegation of the EU, provides assistance and support in spreading information about possibilities and projects of the Erasmus+ program in higher education, professional development, youth, and sports in Ukraine. In response to the war, the European Commission has added new possibilities for supporting and protecting scientists and students of Ukrainian universities. It approved an updated yearly work program for Erasmus+ in 2023. This program focuses on supporting projects that aid in the education and integration of Ukrainian refugees abroad, providing assistance to organizations, students, and professors in Ukraine. The funding is intended for a broad spectrum of projects — from language courses and cultural integration programs to scholarships and general financial support for students and researchers. The program has flexible procedures and exceptional measures for getting grants for educational and academic (credit) mobility, which allows Ukrainian scientists to undergo short-term education/training or internship/professorship, participation in training and volunteer initiatives abroad, etc.

With the beginning of the full-scale invasion, some partners, responding to the EU's decision, reallocated quotas, granting places originally intended for Russian and Belarusian researchers to Ukraine. Additionally, certain Georgian universities voluntarily relinquished their mobility quotas in favor of their Ukrainian counterparts. According to NEO, Erasmus+ currently supports 50 alliances that bring together over 430 universities from all over Europe.

Some **publishers** and online platforms for scientific magazines have also joined the initiatives to support Ukrainian scientists' research and started to give quotas of free publications in their periodicals to Ukrainian scientists as well as free access to their web resources. For instance, Bentham Science Publishers[72], specializing in scientific literature in pharmacology, medicine, technical, and social sciences, addressed the needs of Ukrainian scientists. They waived publication fees for Ukrainian scientists in their 130 peer-reviewed scientific journals indexed in leading databases until the end of 2023. (130 peer-reviewed scientific magazines indexed in the leading databases). They also extended access to digital collections of books and journals for Ukrainian universities and scientific institutions during the same period.

In evaluating the comprehensive spectrum of international support for Ukrainian scientists during the war, data from ScienceForUkraine surveys indicates a shift in preferences among respondents. In the first month following the invasion, scholarships were favored as the primary and quickest form of support. However, as time progressed, the majority of scientists began to prioritize research grants (59.4%) and internships (51.3%). Additionally, over a third of respondents opted for permanent (38.3%) or temporary (37.7%) positions, while just over a quarter (28.0%) preferred scholarships. To upscale research and broaden the relationship with the receiving side, among other support options, scientists chose financial and social support (scholarship, housing, language courses), participation in current research, and work

[71] URL: *https://erasmusplus.org.ua/programa-yes-erazmus/pro-programu/*.
[72] Bentham Science Publishers. URL: *https://www.eurekaselect.com/*.
[73] Maryl, Maciej, Jaroszewicz, Marta, Degtyarova, Iryna, Polishchuk, Yevheniia, Pachocka, Marta, & Wnuk, Magdalena. (2022). Beyond Resilience: Professional Challenges, Preferences, and Plans of Ukrainian Researchers Abroad. Zenodo. URL: *https://doi.org/10.5281/zenodo.7380509.* URL: *https://zenodo.org/record/7380509*.

unrelated to research. The majority (52.7%) favored long-term support (a year or more), while 37.1% preferred midterm support (6-11 months), with only 10.1% opting for short-term solutions[73].

Despite active support and numerous initiatives from international partners, Ukrainian scientists keep facing some obstacles in renewing research:

- positions for vacancies in foreign institutions (according to SAR, #ScienceForUkraine) don't always align with the academic abilities of scientists in Ukraine (there's a disbalance between the offered positions and applicants, so some scientists have to work outside their specialization);
- insufficient proficiency in academic foreign languages;
- lack of experience in preparing grant applications for foreign funding;
- foreign bureaucratic procedures and document preparation timelines, lack of information on administration in other countries,and a prolonged application review process, especially when Ukrainian scientists are in temporary shelter abroad;
- requirement for recognition of education documents;
- legal status of the scientist in the foreign country (e.g., invited employee, person seeking temporary shelter, employee of a Ukrainian research institution, etc.);
- mental state of uncertainty and lack of skills of planning during an emergency.

Based on the analysis of the programs designed to support and provide grants for Ukrainian scientists at risk, it is evident that some initiatives by foreign research institutions were strategically crafted, often aiming more at enhancing their own competitiveness and scientific capabilities than offering unconditional support to Ukrainian colleagues facing challenging circumstances.

In such circumstances, the existence of well-maintained scientific connections and networks with foreign colleagues still plays a decisive part in scientists' choices of refuge and shaping a relatively clear vision of the prospects associated with various offers and their evaluation in short- and long-term research planning.

Nevertheless, volunteer initiatives and steadfast support gained from numerous universities, academic institutes, scientific societies, publishers, and financial institutions worldwide were met with immense gratitude from the Ukrainian scientific community. **The broad support provided by international partners and donors had a decisive significance mitigating the impact of the war on academic and scientific communities in Ukraine.** We may say that the networks created at the beginning of the war laid a foundation for facilitating the integration of Ukrainian scientists into the world community, and in the future, they may become a foundation for the efficient development of more formal modern organizational initiatives with the participation of Ukraine.

4.2. Foreign Academic Mobility of Male Scientists Under Martial Law

Despite the expanded opportunities, only a small number of scientists have managed to move abroad and use the advantages that the global reaction opens before them; the rest remain in Ukraine. Many scientists joined the Territorial Defense Forces or were mobilized to the Armed Forces of Ukraine. Some didn't want to leave their homes and families in Ukraine for a short research trip abroad, others had responsibilities towards children or elderly family members who cannot accompany them. Even among those who could go, many were taken aback by the bureaucratic process of getting long-

term visas. Another important aspect was the **problem of foreign academic mobility of male scientists** under martial law.

Starting from the evening of February 24, 2022, Ukrainian male scientists between the ages of 18 and 60 were subjected to restrictions on their right to freely move outside of Ukraine. To be exact, they could only travel abroad if they met specific criteria outlined in a general list for male citizens of Ukraine within the respective age range. It wasn't until October 2022 that the Ministry of Education and Science of Ukraine issued a letter "About the crossing of the border by employees of educational institutions during the legal regime of martial law" (dated 10/25/2022 #1/12573-22)[74]. The document, in particular, said that the "Ministry of Education and Science of Ukraine, if necessary, addresses the State Border Guard Service of Ukraine about the possibility of going abroad for employees of educational institutions that plan to cross the border to make a trip according to the international obligations of Ukraine. At the same time, we inform the public that the international obligations of Ukraine don't include decisions, acts, protocols, agreements, and other documents, signed or achieved between Ukrainian and foreign educational institutions, or between representatives of these educational institutions". While the letter did not specify the maximum duration of the trip, oral explanations and practical implementation indicated that it was intended for short-term visits ranging from three weeks to a month.

As of June 2023, the situation regarding the ability of Ukrainian male scientists to travel abroad has seen little change compared to 2022: respective scientists can't be mobilized, but no permit to go abroad is provided for this category. According to the third paragraph of the third part of article 23 of the Law of Ukraine "About Preparation to Mobilization and Mobilization", "scientists and scientists-pedagogues, primarily employed at higher and professional pre-higher educational institutions, and research institutions and organizations for at least 0.75 of the workday, are exempt from military service during the special period." However, the rules for crossing the state border by citizens of Ukraine (approved by the Resolution of the Cabinet of Ministers of Ukraine from 01/27/1995 #57) in the current edition from 02/10/2023 state that individuals of male gender aged 16 to 60 cannot cross the border. The existing mechanism for approving the departure of scientists, as described in the mentioned letter from the Ministry of Education and Science of Ukraine, is not formally included in Ukraine's current normative legal acts. In other words, this is a situational initiative created by MES and SBGS to address urgent issues with the departure of separate categories of scientists. Furthermore, this procedure does not guarantee successful border crossing, as final decisions are made by representatives of the SBGS during the crossing. Similar practices involving other governmental bodies exist in various spheres, extending beyond the Ministry of Education and Science.

Despite legislative initiatives introduced in the spring to autumn of 2022 aimed at enabling academic mobility for male scientists under martial law, these initiatives did not progress further.

This situation of uncertainty not only hampers the research opportunities for Ukrainian male scientists but also affects the promotion of Ukraine in the global arena through scientific diplomacy.

The paradox of the situation is that male scientists, despite not having permission to go abroad for more than a month, are exempt from being drafted into the military. This limitation raises practical questions and concerns, especially when it comes

[74] URL: *https://uu.edu.ua/upload/universitet/normativni_documenti/1_12573-22_Pro_peretin_cordonu_NPP_pid_chas_viyni.pdf*.

to participating in prestigious scientific programs. One such example is the Fulbright Visiting Scholar Program, which, in the 2022-2023 and 2023-2024 educational years, restricted male scientists to only five participants for a duration of 3 to 9 months. So, there's an urgent need to clearly regulate the ways of making academic mobility of male scientists possible under martial law by providing permission to cross the border for longer terms.

4.3. Dynamics and Challenges for Academic Mobility of Ukrainian Scientists in the Period of 2022-2023

From the beginning of the full-scale invasion, the biggest flow of "evacuative mobility" abroad was in the spring of 2022. As of autumn 2023, migration abroad had considerably decreased, approaching levels seen before the full-scale invasion. Many scientists opted to become internally displaced persons within the country, choosing to remain in their institutions despite the challenging circumstances. This has provoked **growth in academic mobility inside the country** that was only weakly developed earlier. It was also noted that the management doesn't approve of remote work from abroad, and there are **some documented cases of firing researchers who are abroad** if their internship or research wasn't officially agreed upon beforehand as a collaboration between the institutions.

In the focus group discussions, we asked the participants about:

- the ability to conduct research;
- the attitude of the management towards researchers who found temporary shelter abroad;
- the establishment of new partnerships with foreign colleagues.

During the focus group discussions, we discovered that Ukrainian researchers were quickly adapting to changes, preparing grant applications, and finding partners to conduct research. The strength of the Ukrainian scientific community was evident in its ability to quickly mobilize, respond to challenges, and exhibit flexibility in communication, showcasing immense working capacity. At the same time, researchers who know a foreign language on a sufficient level and have previous experience in collaboration with foreign colleagues turned out to be more successful in forming research partnerships. This points to a significant underestimation of scientific human potential in Ukraine.

Researchers who relocated abroad amid the challenges of displacement described a range of difficulties, including the search for housing, childcare arrangements, and a sense of nostalgia for their homes and families. Despite these challenges, they expressed a strong motivation for professional growth, a commitment to contributing to their home institutions, and a desire to implement innovative practices learned from their experiences abroad. To maintain their connection with their home institutions, these scientists often acknowledged a dual affiliation in their publications, mentioning both their home institution and the institution where they were currently based.. Among interesting practices they are planning to implement in their research institutions and universities, they named informal scientific initiatives, such as a "Night of Scientific Research", "Scientific Party", and "Scientific Caffe". They also noted the balanced workload and work-life balance

observed in their Western colleagues, which wasn't possible in Ukrainian realities even before the war. Among factors that influence decision-making on location were named: 1) ability to research topics related to Ukraine; 2) conditions and infrastructure for researchers' professional growth; and 3) research funding. At the same time, they highlighted barriers to returning to Ukraine: 1) it's impossible to come back because of occupation or active combat; 2) children attending a local school; 3) employment of their partner[75], and family ties.

Consolidating the scientific community and support allowed Ukrainian scientists to research in foreign research institutions in a safe place. However, over 80% of scientists stayed in Ukraine. The existing limitations in Ukrainian laws, particularly those affecting academic freedom for male scientists under martial law, hindered their ability to fully leverage international support.

The researchers who found temporary shelter abroad are forming a new generation of Ukrainian scientific diaspora, so it's important to establish mechanisms for professional networking between their foreign institutions and Ukrainian counterparts to develop institutional collaboration, language and intercultural competencies in young scientists.

[75] This remark isn't relevant for the focus of this research, but we paid attention to it. Among the reasons for relocation, employment of the husband was noted, but employment of the wife was not. This may both be connected with the gender imbalanced sample of the respondents, or with the traditional conservative role for a woman.

Chapter V. Prospects for Preserving Human Resources of Ukrainian Science During Countering Russian Full-Scale Invasion

The war is a period of immense trial for a nation that causes qualitative changes in all social groups. For the Ukrainian scientific community, the full-scale Russian invasion created both threats and new horizons for development. To preserve scientific potential, it's important to focus on: 1) preserving people, scientists as a source of new ideas, new knowledge, and inventions; 2) preserving scientific schools as a succession of scientific traditions and ensure the ongoing continuation of research in the same direction in case of the demise of the founder of a field; 3) preserving research topics as a basis for reviving scientific traditions and forming a circle of followers.

The war has led to irreversible losses in science: deaths of researchers, deterioration of their health and psycho-emotional state, destruction of infrastructure, and forced migration. This poses a threat to reproduction of Ukraine's scientific potential, considering the risks of some migrants not returning, decreased efficiency of scientists, and weaker training of the younger generation. However, in overcoming these challenges, there is a strengthening of the national identity of the population, an activation of its social responsibility, development of digital skills, and formation of new cultural narratives. All this creates new possibilities for quality changes and a quantum leap on a new level of development in Ukrainian science. This thesis can be confirmed by the results of a study by Ukrainian scientists of the Institute of Industrial Economics of the National Academy of Sciences of Ukraine to identify threats and opportunities for the Ukrainian scientific community in the context of war (Fig. 7)[76]

[76] Новікова О., Залознова Ю., Азьмук Н. Відновлення людського капіталу України у післявоєнний період з використанням переваг цифровізації, Журнал європейської економіки, Том 21, № 4 (83), Жовтень-Грудень 2022, с. 407-427 URL:https://jeej.wunu.edu.ua/index.php/ukjee/article/view/1619/1622.

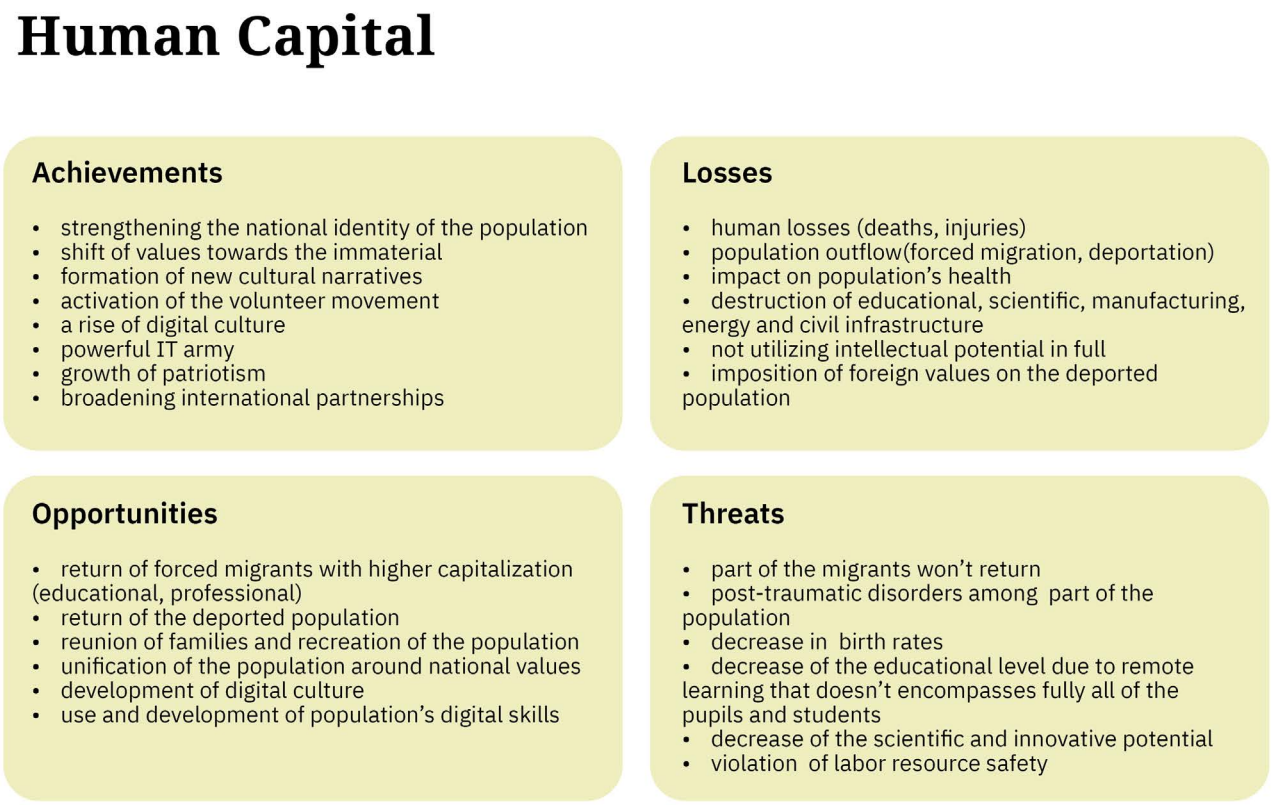


Fig. 7 Systematization of achievements, losses, potential opportunities, and threats to the development of human capital. Authors: Olga Novikova, Yuliya Zaloznova, Nadiya Azmuk, 2022

Based on the analysis of threats and opportunities that arose during the war for the Ukrainian scientific community, measures for preserving scientific human resources under optimistic and base scenarios were developed. Main assumptions of the optimistic scenario: 1) Safety; the war will end in 2024 (prediction of the then Defence Minister of Ukraine O. Rieznikov[77], head of the Main Directorate of Intelligence K. Budanov, statement by President of Ukraine that the war will end before the USA elections); 2) Funding opportunities for scientific researches in Ukraine will be expanding; 3) economics recovery will be based on innovative principles; 4) research infrastructure will be renewed; 5) ecology will not deteriorate, military actions will not lead to additional technogenic accidents. Main assumptions of the basic scenario: 1) The war will last five more years[78] (Institute for the Study of War); 2) Funding opportunities for scientific researches in Ukraine will not be reduced; 3) The macroeconomic situation of Ukraine will not worsen; 4) Research infrastructure will not be significantly damaged as a result of military actions; 5) Ecology will not deteriorate, the impact of military actions on the environmental situation will be locally limited.

[77] When will the war end? The promises of Budanov, Rieznikov, Arestovych - the boldest statements. URL: *https://glavcom.ua/country/politics/dva-tri-tizhni-ta-zvilnennja-krimu-u-travni-prohnozi-shchodo-zakinchennja-vijni--949817.html*.

[78] Eurocontrol predicts limitations on flights over Ukraine till 2029. URL: *https://www.eurocontrol.int/sites/default/files/2023-03/eurocontrol-seven-year-forecast-2023-2029-spring-2023.pdf*, *https://ig.ft.com/russias-war-in-ukraine/*

Table 3. The list of measures for preserving scientific potential under optimistic and base scenarios.

Measures for	Basic Scenario	Optimistic Scenario
Preserving scientists (people)	Safety and medical trainings . Equipping covers and bomb shelters. Investment in developing scientific human potential (work with pupils, JAS). Conducting events to popularize science (like "Night of Scientific Research", etc.) Development and improvement of language, digital, cross-cultural and communication competencies in scientists. Development of normative legal regulation to facilitate academic mobility for male scientists. Mental health support for scientists at risk for conducting research.	Measures for supporting the return of scientists from AFU to scientific work. Programs for adaptation of scientists after service in AFU. Measures to form inflows of returning migration of scientists to Ukraine (for example, grant support of scientists who return, institutional support, creation of laboratories, etc.). Balancing the workload for scientists, especially in universities, depending on their scientific results. Flexible administration of remote work
Preserving research groups	Informing and promoting research results of Ukrainian scientists among international professional communities. Formation of a network of foreign research hubs. Development and institutionalizing a network of the Ukrainian scientific diaspora. Establishing communication channels and setting up efficient communication between the Ukrainian scientific diaspora abroad and research groups in Ukraine. Creating a mechanism of engagement of Ukrainian research groups to work in international scientific collectives (scientific outsourcing).	Creation of initiatives and foundations to support leading research and research groups. Informing about grant support opportunities for research, research groups, and programs for learning and improving skills in preparing grant applications. A digital platform that will present Ukrainian research groups, their research topics, and results will be located.
Preserving scientific topics of research	Creation of a network structure to conduct research by opening remote positions for scientists who cannot leave Ukraine. Improvement of digital mechanisms for preserving research results (creation of a national repository of academic texts and ensuring its regular updates).	Creation of international platforms for presenting research results of Ukrainian scientists. Creation of short-term scholarships for foreign young scientists who research Ukrainian subjects or collaborate with Ukrainian research groups. Creation of a mechanism of national recognition (a national prize) for research results in different fields of science.

The measures offered in Table 3 may be transformed into decisions by the leaders of scientific institutions and university rectors or into regulatory and legal acts, as indicated in the "Recommendations" section.

Conclusions

Based on the analyzed material, we may single out a couple of important theses:

1. Studying global experience has shown the variability and vulnerability of international connections related to safety issues, evidenced by the outbreak of new armed conflicts in unpredictable places. Human rights and academic freedom are violated during such conflicts, which prevents scientists from conducting research. This causes migrational waves of scientists to safe places and their departure from the profession.. To preserve scientific achievements and ensure scientific progress, the global scientific community unites to support scientists at risk by creating global initiatives and initiatives on the national level or the level of individual universities.
2. Ukraine has no unified mechanism and accounting system for losses of scientific human resources. The tendency of losing human potential in Ukrainian science emerged earlier due to the increase in the median age of scientists, but this issue became dramatically more urgent due to the full-scale Russian invasion.
3. The Ukrainian scientific community consists of such groups: 1) scientists who stayed in Ukraine; 2) scientists who serve in AFU; 3) scientists who have found temporary refuge outside Ukraine; 4) scientists demobilized from AFU. Scientists serving in AFU were forced to put their professional lives on hold; however, if a scientist belongs to a different group, it doesn't mean they can conduct research efficiently. should be noted that they are at risk independent of a scientist belonging to any of these groups. However, the types of risks differ, and measures to overcome them and mechanisms for preserving scientists from different groups will differ.
4. The Ukrainian scientific community has demonstrated resilience in producing research results, but the full-scale war has negatively impacted their overall quantity. Scientific collaborations of Ukrainian scientists have also changed.
5. Scientists who found temporary refuge outside Ukraine and continue to conduct research are forming a new generation of Ukrainian scientific diaspora which can lead to efficient scientific diplomacy and contribute to the development of international collaborations.
6. As a result of the full-scale invasion, Ukrainian scientists have faced risks

associated with being in the zone of armed conflict (threat to their lives and well-being, violations of human rights, etc.), the imperfection of the Ukrainian laws that limit the academic freedom of male scientists, in particular, the limit on going abroad that doesn't give the possibility to participate in scientific conferences, communicate with foreign colleagues, make experiments on unique equipment, etc.; closure of their research topics due to changes in priorities, forced relocation, and discontinuation of funding.

7. Ukrainian scientists have received unprecedented international support and opportunities to conduct research in safe places outside Ukraine. However, more than 80% of scientists remained in Ukraine due to both family circumstances and imperfections of state regulations. Scientists who have sufficient knowledge of a foreign language and had previous experience of cooperation with foreign colleagues were able to fully utilize international grant support for scientific research.
8. The majority of assistance programs from international partners were aimed at assisting scientists who moved abroad. However, nobody expected that most Ukrainian scientists would stay in Ukraine, so such remote support programs appeared later.
9. Ukrainian institutions exert pressure on employees by firing them because of refugee status, which leads to the loss of connections between researchers and professors and their institutions and putting on hold/terminating research in their professional topic.

Recommendations

1. For executives of research institutions and universities:

- Implement best practices to assist and support Ukrainian scientists abroad, understanding their role as ambassadors who can build bridges between Ukrainian and foreign educational and research institutions for potential academic collaborations, exchange programs, and partner research projects[79].
- Create programs and conditions for professional development (for example, writing grant applications, negotiations, cultural peculiarities), learning foreign languages, and exchanging experiences with foreign colleagues.
- Upscaling scientific communication: widespread information about international collaboration programs and grants, development of communication culture with the international scientific environment, overcoming language barriers, and individual mobility in science and innovations. **Advantages**: development of the scientific potential of scientists; Broadening the network of representative platforms for conducting scientific activities and presenting research results; minimization of the presence and potential influence of the aggressor country in Ukrainian science; additional sources of funding and gradual distancing from the research funding by Russian organizations; increasing the capitalization (educational, professional) of scientists; development of shared scientific topics with foreign scientists; ability to use research infrastructure abroad remotely.
- Facilitate the optimization of remote work and flexible working hours for scientists and introduce digital documentation. Reduce scientists-lecturers' workload depending on their scientific achievements (balance auditorium, methodological, and scientific workload on professors).
- Introduce special scholarships to support scientists' research/preserve scientific schools. Example: Bizhan Sharapov scholarship for young researchers.
- Develop and implement support programs for scientists on the frontline and programs for their return to science. Scientists who continue working, grow and develop professionally, while time is seemingly standing still for those on the frontline.
- Ensure the physical preservation of employees of the institution by optimizing its structure according to safety systems for wartime (modern equipment of covers, bomb shelters, efficient evacuation plans), and also conducting basic safety training and tactical medicine briefings for scientists.
- Change policies on the employment of scientists who have found temporary refuge abroad; in particular, avoiding forced dismissal due to a scientist's stay

[79] Universities during the war: from the educational institution to social mission *https://ekmair.ukma.edu.ua/server/api/core/bitstreams/c20b0b6d-be52-4b0b-8544-2fbdbc4850dd/content* (retreived on 09/26/2023)

abroad. Strive to preserve scientific human resources in departments and research centers. Ensure preserving professional contacts of the scientists with Ukraine while they are continuing research abroad, which will allow establishing important ties and learning new methods of research that may help Ukraine turn into a more modern and internationally integrated producer of science (also, this way there will be higher chance that the researchers will come back and will implement their new knowledge in Ukraine).

2. For governmental bodies and local authorities in the country:

- Finish the work of systematically accounting data on losses of scientists in wartime (scale of destroyed and damaged research infrastructure, relocation of scientists around the country and outside of it, dispersing of scientific groups and scientific partnerships and creation of new ones) started by MES, agree upon the mechanisms of such accounting with all the stakeholders (MES, NASU, research institutions, and universities) that are subordinate to different ministries. Introduce additional systematization of data by age and gender of the scientists. Advantages: the ability to make balanced decisions and track their efficiency, formulation of clear requests for assistance from international donors, and creation of a foundation for post-war recovery.
- Initiate the practice of publishing systematized data on losses of scientists in wartime.
- Regulate ways of implementing foreign academic mobility of male scientists under martial law by providing the ability to cross the border, not just for short-term trips.
- Create possibilities for scientists in AFU to continue research by engaging them in scientific developments in the defense sphere or letting them serve in appropriate institutions (for example, the Center for Innovation and Defence Technology).
- Reduce the workload on scientists, research institutions, and universities by simplifying the reporting procedures and project support. Transition the interactions with authorities into a digital format, avoid repeatedly providing information, etc.
- MES should implement adequate policy on requirements of reporting and academic bureaucracy and introduce more favorable legal conditions so that researchers receive their salaries and implement projects even after returning to Ukraine.
- Develop initiatives for introducing bilateral research projects from MES (for example, Polish-Ukrainian research programs or Ukrainian-German collaboration).
- Regulate legal framework regarding the possibility for Ukrainian scientists to hold remote scientific positions in institutions of partner countries.

3. For active citizens and non-profits

- Spread the information to the Ukrainian audience about the issue of preserving Ukrainian scientists during the war, through the media, on social media, on websites of partners and institutions they collaborate with.
- Convey information to the international audience by engaging the media and social networks, publishing on projects' websites, and by using international forums and platforms for discussion (like, for example, the European Strategy Forum on Research Infrastructure, ESFRI).
- Facilitate the strengthening of the influence of extant foreign scientific institutions and use of the potential of the Ukrainian scientific diaspora that is being formed, in particular, by opening new scientific hubs/platforms.

4. For global international institutions:

- Use Ukrainian experience to systematize and synchronize standards, values, basic principles, and overall policies for responding to crisis situations that may

threaten losses of human resources in science (defining clear responsibilities, mechanisms of coordination, and rapid reacting in an emergency).
- Create international support networks for scientists at risk.
- Within the framework of international partnership and mutual assistance policy, develop legal and financial mechanisms for institutions of different countries to grant remote research positions to scientists affected by extreme crisis phenomena (wars, armed conflicts, natural disasters, etc.) to enable researchers who cannot leave the risk zone to continue research.

5. For international partners:

- For international donor organizations — introduce support programs for scientists-professors to compensate for their factual workload connected with work under martial law.
- Broaden programs to support setting up internal points of invincibility* for universities and introduce them to all institutions.
- Upscale grant funding for scientific developments by Ukrainian scientists.
- Establish scientific centers/affiliations based on the Ukrainian scientific diaspora to preserve research with Ukrainian topics.
- Actively engage Ukrainian researchers in international chains of creation of scientific products through efficient mechanisms of remote work and outsourcing (developing and holding tandem courses, co-supervision with postgraduates, execution of research/parts of research ordered by foreign institutions).
- Reformat parts of programs to support scientists who remain in Ukraine or have returned from abroad (support of electric power independence, financial support within the framework of virtual mobility and remote grants/scholarships, short-term offline events to forge networks of collaboration with foreign colleagues, etc.).
- Change focus towards flexible support programs, taking into account the changeable crisis phases ("state of emergency" — "long, lasting crisis" — "phase of revival and rebuilding").
- Introduce integration programs for forming international research groups where Ukrainian scientists who know the foreign language (English or the language of the receiving country) and have appropriate professional competencies will be included in positions of leaders of research groups. **Advantage**: this is an efficient way for Ukrainians who do not yet speak the language of the host country to adapt, enabling effective communication between them and foreign employers.
- At the level of scientific and educational institutions, establish institutional mechanisms for introducing remote research positions to scientists who are at risk and cannot take advantage of academic mobility or grant programs that demand going abroad.

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APPENDICES

Appendix 1. Armed conflicts during the 20th–21st century and their threats to the scientific community

Armed conflict	Description	Risks and threats for scientists (scientists at risk)
WWI (1914-1918)	The conflict between the Entente (France, Great Britain, Russia, and others) and the Central States (Germany, Austro-Hungary, the Ottoman Empire). It led to significant human losses among the civilians. It is estimated that the number of civilian victims is around 6-13 million people. Among them were scientists and researchers, but it's hard to establish accurate data on their numbers.	A threat to lives and well-being due to armed conflict. For the first time, scientists were perceived as vital for the war effort; they were working on scientific problems from aviation and identification of submarines to chemical weapons[80].
WWII (1939-1945)	A global armed conflict, in which over 60 countries were involved. Over 100 million people took part in combat directly. The number of victims is estimated to be 70-85 million people. As a consequence of numerous mass murders and crimes against humanity (including the Holocaust, strategic carpet bombardments, and the use of nuclear weapons) many scientists died, and many were forced to move to safe places. **At the same time, many scientists took part in the military research.**	A threat to the lives and well-being of scientists due to armed conflict, crimes against humanity, and genocide. The **Council for At-Risk Academics** was created, which helped rescue prominent scientists from the Nazis[81]. The universities of Austria and Germany couldn't overcome the loss of research potential from the emigration of Jewish scientists for 50 years[82,83].
Cold War (1947-1991)	Geopolitical antagonism between the USA and USSR. It didn't involve open warfare but influenced world politics and the arms race, the intensive development of military technology, and the space industry[84].	**A significant limitation of academic freedom** and the cessation of information exchange between the East and the West. Many scientists were suspected spies; some were under law enforcement surveillance or were forced to relocate or fired for political reasons.

[80] Emery, T., White, J. B., Sitrin, S. S. and C., & Gerstein, B. M. and J. (2018, November 12). How World War I became the first modern war of science. POLITICO Magazine. URL: *https://www.politico.com/magazine/story/2018/11/12/world-war-one-science-222410/#:~:text=,War%20would*.
[81] CARA (the Council for At-Risk Academics) Homepage. Home. (n.d.). URL: *https://www.cara.ngo/who-we-are/our-history*.
[82] Waldinger, F. (2012). Peer Effects in Science: Evidence from the Dismissal of Scientists in Nazi Germany. The Review of Economic Studies, 79(2), 838–861. URL: *http://www.jstor.org/stable/23261352*.
[83] Becker, Sascha O & Lindenthal, Volker & Mukand, Sharun & Waldinger, Fabian, 2021. Scholars at Risk: Academic Networks and High-Skilled Emigration from Nazi Germany. The Warwick Economics Research Paper Series (TWERPS) 1330, University of Warwick, Department of Economics.
[84] Baillie, K. U. (2021, January 11). The interplay between scientific progress and violence in Modern War. Phys.org. URL: *https://phys.org/news/2021-01-interplay-scientific-violence-modern-war.html*.

Armed conflict	Description	Risks and threats for scientists (scientists at risk)
War in Vietnam (1955-1975)	An armed conflict between North and South Vietnam, supported by, respectively, the USSR and the USA. The number of victims is estimated to be around 2-3 million people. In the Vietnam Republic, as of 1975, there were 83 thousand amputees, 30 thousand blind, and 10 thousand deaf people because of the war.	The "Yellow Berets" program by the National Institutes of Health for training medical school graduates in the USA increased the probability that its participants wouldn't limit themselves to medical practice but would prefer to conduct research[85].
Occupation of Cyprus by Turkey in 1974	It's not a bloody armed conflict in its character, but scientists from Northern Cyprus found themselves isolated from the international scientific community.	Significant limits on academic freedom and isolation of researchers caused a "brain drain".
Iran-Iraq War (1980-1988)	An armed conflict between Iran and Iraq that led to the deaths of over 1 million people. During the war, Iraq used chemical weapons.	Large human losses, a large number of displaced persons, and a mass exodus of scientists from the country.
Balkan (Yugoslavia) Wars) (1991-1999)	A series of armed conflicts that took place on the territory of the former Yugoslavian republics (Croatia, Bosnia and Herzegovina, and Kosovo). The conflicts changed borders in the region and led to around 140 thousand deaths.	The publication activity of scientists in the territories where the active phase of the armed conflict took place reduced significantly.
War in Iraq (2003-2011)	Invasion of the USA and allies to Iraq to remove Saddam Hussein from power. It happened without the UN mandate. Around 150-200 thousand people died.	Murders, kidnappings, and persecutions of researchers. Till January 2007, around 300 scientists were killed, including doctors of sciences who worked in Iraqi governmental ministries and also as administrators of universities[86]. Over 3,250 scientists left the country from February to August 2006[87]. 84% of the Iraqi universities' infrastructure was burned, robbed, or seriously ruined in some form.
War in Syria (2011 — present)	The civil war in Syria has been lasting since 2011 and caused significant humanitarian, economic, and political challenges. Various internal and international sides take part in the conflict, including the Syrian government, the rebels, Muslim extremists, and others. The number of victims is in the hundreds of thousands.	Governmental persecution of scientists. Appearance of scientific refugees: scientists who needed help outside of their country[88]. The campus of the Aleppo University was destroyed (more than 80 people died).

[85] Pierre Azoulay, Wesley H. Greenblatt, Misty L. Heggeness, Long-term effects from early exposure to research: Evidence from the NIH "Yellow Berets". Research Policy, Volume 50, Issue 9, 2021. URL: *https://www.sciencedirect.com/science/article/abs/pii/S0048733321001311*.
[86] Scientists become targets in Iraq. Nature 441, 1036–1037 (2006). URL: *https://doi.org/10.1038/4411036a; https://www.nature.com/articles/4411036a*.
[87] Baghdad Bulletin: Death Stalks the Campus. URL: *https://web.archive.org/web/20070304192518/http://www.time.com/time/world/article/0,8599,1553793,00.html*.
[88] Stoepler TM, Creswell JE, Anthis NJ, et al. The role of science diplomacy in international crises: Syria as a case study. Sci Dipl. 2017;6. URL: *https://www.sciencediplomacy.org/article/2017/role-science-diplomacy-in-international-crises-syria-case-study*. Accessed 1 March 2023.

Armed conflict	Description	Risks and threats for scientists (scientists at risk)
The Russian War against Ukraine (2014 — present)	The Russian war against Ukraine started in 2014 and is lasting in the present.	The appearance of scientists-refugees, destruction of scientific infrastructure, mobilization of scientists in AFU. Large-scale missile attacks on Ukrainian territory, partial or total destruction of cultural and historical heritage monuments, universities, and scientific institutions.
Taliban coming to power in Afghanistan (2021 — present)	The Taliban coming to power in Afghanistan happened in August 2021. In this conflict, the sides were the Afghan government and rebels from the Taliban, who gained victory.	Limits of academic liberty, the closing of universities. The Taliban coming to power has negated two decades of progress in higher education, as the new de-facto government introduced policies and practices that systematically rob female students and scientists of the right to higher education. The universities controlled by the Taliban gave orders about the arbitrary firing of professors and administrative staff, in particular, on an ethnic basis, and the Taliban police arrested scientists who criticized the new government[89].
War in Israel (10/07/2023)	The Hamas has massively attacked the Israeli territory with missiles, crossed the border of the Gaza Strip, and occupied seven districts in the South. As of 10/11/2023, it is known about 700 dead on both sides of the conflict.	Physical losses of scientists on both sides. Israel: cessation of international sponsorship for scientific research, return of foreign postgraduates to their countries (often on the demand of the country of citizenship), scientists joining the military. Over 50 students, professors, and employees of the Ben-Gurion University of Negev (41 km to the Gaza border) were murdered or kidnapped as a consequence of the Hamas attack[90]. **Gaza**: before the current war, a defensive blockade complicated getting funding for Palestine researchers, limited academic mobility, and import of scientific equipment. Two big institutions — the Islamic University of Gaza and the Al-Azhar University — were significantly damaged, and it's hard to accurately estimate losses in scientific and educational infrastructure because of the total destruction.

[89] Mallapaty S. Afghanistan's terrified scientists predict huge research losses. Nature. 2021;597:15–16. URL: *https://doi.org/10.1038/d41586-021-02353-5*.

[90] Israel-Hamas war sends shock waves through scientific community. *https://www.science.org/content/article/israel-hamas-war-sends-shock-waves-through-scientific-community*.

Appendix 2. Comparison of indices of publishing activity in different fields of knowledge (based on the Scopus database)

№	Fields of knowledge	Publishing period 2020-2021	Fields of knowledge	Publishing period 2022-2023
1	Engineering	11 703	Engineering	7 878
2	Computer Science	9 065	Physics and Astronomy	5 741
3	Physics and Astronomy	8 851	Computer Science	5 157
4	Medicine	6 740	Medicine	4 714
5	Materials Science	6 573	Materials Science	4 452
6	Mathematics	5 135	Social Sciences	3 805
7	Social Sciences	4 953	Mathematics	3 321
8	Energy	3 816	Environmental Science	2 726
9	Environmental Science	3 634	Agricultural and Biological Sciences	2 492
10	Chemistry	3 180	Chemistry	2 477
11	Agricultural and Biological Sciences	3 104	Energy	2 342
12	Earth and Planetary Sciences	2 954	Biochemistry, Genetics and Molecular Biology	2 150
13	Biochemistry, Genetics and Molecular Biology	2 785	Business, Management and Accounting	1 745
14	Business, Management and Accounting	2 521	Earth and Planetary Sciences	1 726
15	Arts and Humanities	2 247	Economics, Econometrics and Finance	1 521

Appendix 3. The form for research on the motivation of scientists for relocation/return to Ukraine during the focus group

1. Motives for Relocation Abroad: Please rate the factors that influenced your decision to relocate for research on a scale from 0 to 5, where 0 means 'no influence' and 5 indicates a 'very strong influence'.

1. Safety: the ability to be safe during the Russian war against Ukraine (physical safety is meant here — to be outside of the range of shelling, occupation, or air-raid sirens).
2. Professional development: possibilities for professional development, improvement of the scientific reputation, and better conditions for career development.
3. Welfare: more comfortable living conditions, conditions for conducting research, keeping good health, less stress and psycho-emotional load.
4. Family: better education quality for children, access to kindergartens and schools, and better opportunities for development and raising a child.
5. Research funding: availability of grants, scholarships, or funding programs that support research projects.
6. Research infrastructure: access to unique equipment, information resources, and research environment (unique flora, geography, etc.).
7. Opportunities for integration into international research groups that work on cutting-edge research. The search for answers to global challenges is impossible without global interdisciplinary collaboration and coordination.
8. Reliable intellectual property protection, ensuring compliance with labor rights, etc.
9. Possibilities for implementing innovative developments.

2. Barriers to relocation abroad. Please rate the factors that can hinder your decision about relocating abroad for research on a scale from 0 to 5 where 0 means 'no influence' and 5 indicates a 'very strong influence'.

1. Language and cultural differences that would hinder efficient communication and collaboration with colleagues.
2. Bureaucratic and formal limitations: unavailability of information about relocation, complicated procedures to get a visa and a work permit abroad, complicated procedures of evaluation and recognition of qualification, and other bureaucratic procedures.
3. Financial expenses for living that are disproportionally high for yourself (and your family), renting an apartment, etc.
4. Institutional and administrative limitations (instances of discrimination against migrants/Ukrainians, high competition).
5. Welfare: harder living conditions, the need to constantly look for a home, more stress, much more work is needed, and it's necessary to learn new things, life balance would be upset.

6. Availability of established social and professional connections: a familiar circle of friends, like-minded people, and colleagues. Availability of established professional connections in your country.
7. Social guarantees: better access to healthcare, guaranteed benefits, pension system, complications in looking for a school for your child, the necessity to care for ill relatives, availability of childcare in your country, etc.

3. Motives for return to your country for research. What would prompt you to return to your country after a long-term stay abroad? (0 — doesn't influence at all; 5 — very strong influence)

1. Safety: improved safety situation, more confidence and stability.
2. Professional development: availability of new interesting career opportunities in Ukraine, including an academic position or leadership in research.
3. Welfare: stability and better life quality for me, my children, and my family.
4. Family: access to kindergartens, schools, childcare, and the necessity to care for relatives.
5. Research funding and research infrastructure: availability of attractive grants for research in Ukraine, improvement of infrastructure to conduct research, availability of unique conditions or laboratory equipment.
6. Opportunity to bring cutting-edge experience and become a pioneer in your field in Ukraine, and also the ability to use your knowledge, professional connections, and experience gained abroad to conduct research in Ukraine, promote national priorities, develop Ukrainian scientific institutions, and have a positive influence on Ukrainian society.
7. Social guarantees: better access to healthcare, guaranteed benefits, pension system, etc.
8. Patriotism: the feeling of duty, responsibility for the fate of your country, desire to contribute to the development of science in Ukraine, and a possibility to join forming agenda for Ukrainian science.

4. Research on barriers to returning to Ukraine for research. What would be a barrier for you to return to Ukraine after conducting research abroad? (0 — doesn't influence at all; 5 — very strong influence)

1. Safety: instability of the safety situation, loss of home, uncertainty of the future, danger for the child that depends on you.
2. Professional development: career and research opportunities limitations, including the absence of academic positions and work conditions. Return to Ukraine would mean finishing research in a foreign institution and uncertainty with employment in Ukraine.
3. Welfare: lower wages, worse legal and social protections, lower availability of high-quality education for your child.
4. Family: the zone of vital interests of family members (children, husband) is outside of Ukraine, including the inability to relocate your family or disturbance in your children's education.
5. Research funding and research infrastructure: the absence of funding for research, limited access to state-of-the-art laboratories, equipment, research infrastructure, and communications with colleagues abroad.
6. Social guarantees: lower access to healthcare, pension system, etc.
7. Administrative and bureaucratic obstacles, the spread of discriminative practices: the necessity to renew professional networks and collaboration in Ukraine. The scientific achievements abroad would be negated or wouldn't be fully acknowledged in Ukraine. The need to do formal procedures for acknowledgement of the qualification in Ukraine and renewing the scientific reputation in the Ukrainian scientific community.
8. Biased and discriminative attitude to the relocated, differences in research culture, and research practice in Ukraine and abroad.

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